Informatica®

Informatica® Proactive Monitoring for PowerCenter

3.0.2

Operations User Guide

Informatica Proactive Monitoring for PowerCenter Operations User Guide
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Preface

The *Proactive Monitoring for PowerCenter Operations User Guide* provides information about the Proactive Monitoring components.

It describes the environment and the monitoring components. It also contains reference material that describes components such as rules, watchlists, and topics. This guide is written for users who configure monitoring for the PowerCenter environment.

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CHAPTER 1

Proactive Monitoring for PowerCenter Operations

This chapter includes the following topics:

- Introduction, 9
- Solution Components, 10
- Solution Usage, 13
- Proactive Monitoring Management Console, 14

Introduction

Proactive Monitoring for PowerCenter Operations provides advanced monitoring capabilities for PowerCenter.

An Informatica domain may contain numerous repository services, databases, integration services, and Web Services Hub running on multiple physical or virtual machines.

The Proactive Monitoring solution collects data from the PowerCenter services and their host machines at regular intervals, checks for anomalies in the processing of workflows and sessions and alerts appropriate users. The alert messages contain the required contextual information, such as the session name, workflow name, and cause of the alert, thus enabling Informatica domain administrators, developers, or architects to take corrective action effectively.

The solution contains rules that perform a wide range of checks against the PowerCenter runtime and generates alerts whenever there is a deviation. A simple example is to alert users when a session completes successfully, but loads zero rows in the target system. A complex example is to alert users when a session fails to run, after it was changed by developer. You can also create and change rules based on your business requirements.

A single installation of the solution monitors a single domain, and the multiple services and hosts in the domain. Most alerts generated by the Operations monitoring solution include CPU and memory consumption details of the hosts where the workflow or session is executed.

Proactive monitoring solution provides targeted alerting capabilities. The solution contains predefined personas who receive alerts specific to their function.

The Proactive Monitoring solution contains the following predefined personas:

- pcadmin. Any user who ensures proper functioning of Informatica domains, and PowerCenter Integration Service, Repository Service, and other services.
- apparchitect. Any user who is responsible for the logic of PowerCenter mappings, mapplets, transforms, sources, and targets.
- dataarchitect. Any user who oversees the data movement with in PowerCenter.
- itsecurity. Any user responsible for dealing with IT security issues, such as sensitive data and malware.
- pcmonitor. Any user who tracks PowerCenter performance. By default, the pcmonitor persona receives all alerts.

In a continuous monitoring system, the solution can detect the same anomaly multiple times and send the same alert multiple times. To avoid such a situation, the solution provides the snooze feature. You can use the snooze feature to set a time period before which the solution does not generate any alert on the same anomaly.

For example, a developer updates a PowerCenter object that results in a session processing failure. This could be a planned development activity where the architects would want to stop receiving alerts for a specified period of time.

You can use the Reports dashboard in the Proactive Monitoring solution to monitor the status of application services and hosts in a domain. You can view the details of execution failures that occurred over a period of time.

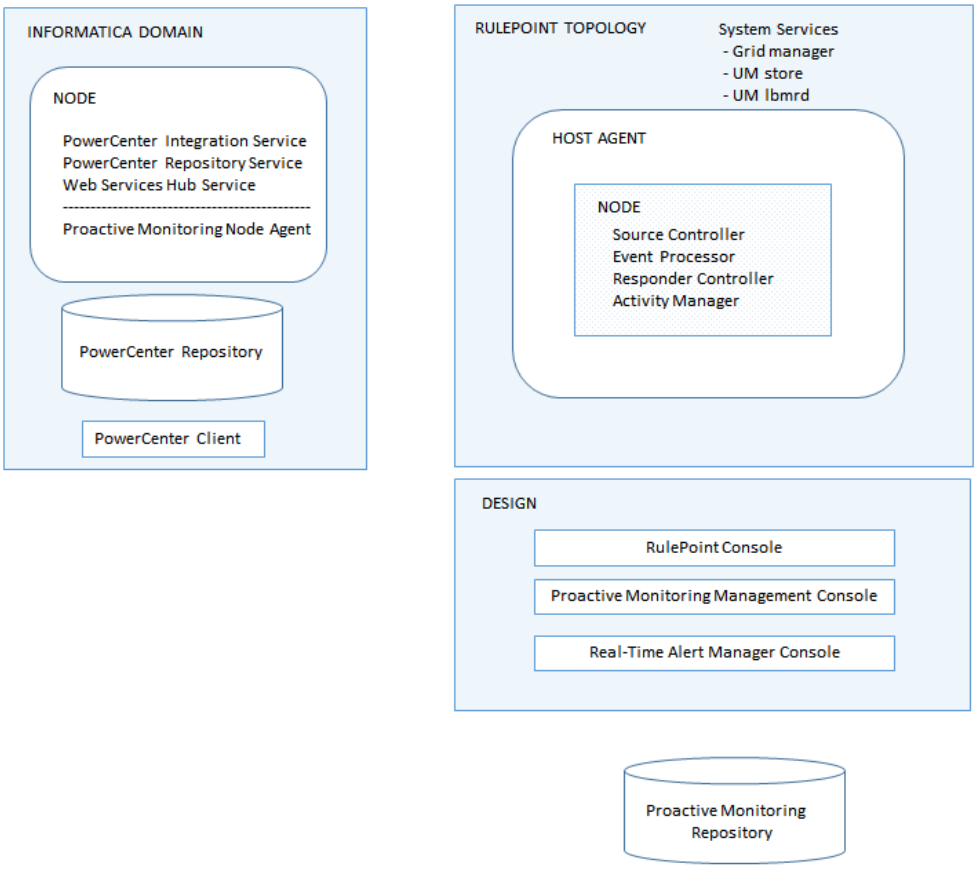
Use the Manage Objects tab to manage objects configured for monitoring PowerCenter. You can manage rules, sources, folders, watchlists, and topics. You can also view the monitored events and activations.

Solution Components

The Proactive Monitoring solution monitors both large and small Informatica domain configurations.

The Proactive Monitoring solution requires the infrastructure for continuous data collection, continuous processing, analysis, and continuous alerting. Informatica RulePoint provides such an infrastructure and programming model.

The following image shows the components within the Proactive Monitoring solution:



Informatica RulePoint

Informatica RulePoint is a general purpose Complex Event Processing server. You can build an application on RulePoint to process large amounts of data in real time, detect anomalies, and take action. Informatica RulePoint provides a programming model to build such applications.

The RulePoint programming model consists of the following primary objects:

- Sources. Sources connect to external systems to fetch data. The sources convert the fetched data into events. The events are published on topics.
- Rules. Rules process events on topics. You can use the rule processing language, DRQL to create rules. The rule definition includes information about the topics and the number of events you want RulePoint to process, the conditions to check, and the response to generate when there is a deviation. As part of overall rule processing, the rule might call out functions called analytics. Rules also rely on watchlists that act as reference data sets during rule processing.
  RulePoint also provides an easy way to create rules using templates. Templates are abstract rules that could become a rule when you provide all its parameters.
- Responders. Responders dispatch alerts to external systems. When a rule condition is matched, the rule processing engine creates a response that is sent to the specified external system using a conduit called responder.

RulePoint consists of the following supporting objects that you reference in the primary objects:

- Topics. Logically group events into a group. A topic describes the types and properties of events coming into the system.
- Connections. Connect RulePoint objects, such as sources, analytics, and responders to the target database.
- Response. Define how you want RulePoint to respond if the event matches the conditions defined in the rule.
- Analytics. Analyze data within a system and implements a data processing function.
- Watchlists. Contain the items that you store as a single object with a unique name that you define. The rule uses this name so that it can use the data stored in the object.
- Templates. Enable users to easily create new rules. A template includes a rule statement that contains substitution parameters and instructional text to define those parameters.

For more information about the objects, see the *RulePoint User Guide*.

Proactive Monitoring for PowerCenter ships with a set of sources, topics, rules, templates, analytics, watchlists, and responders that are specific for the operational monitoring of PowerCenter.

Object States

Objects are in Draft state when you create the objects, you never deploy the objects, or when you undeploy objects. You need to deploy the objects to the application services for rule processing to begin. In the default topology, when you deploy the objects, the grid manager deploys the sources along with the supporting objects to the source controller, responders and supporting objects to the responder controller, and rules and supporting objects to the event processor. When you successfully deploy the primary objects and their supporting objects, the objects are in Deployed state. After you deploy the objects, the source controller begins to fetch events, the rule processor processes events, and the responder controller dispatches alerts.

If you need to change the object properties, undeploy the objects. When you undeploy primary objects, all secondary objects associated with the primary objects are also undeployed. After a successful undeploy, the state of the objects changes to Draft. After you complete the changes, you can deploy the objects again. When you edit and save the objects, the state of the objects changes to Needs_Deployment state. You need to deploy the objects again. For more information about deployment and the state of RulePoint objects, see the "Managing Deployment" chapter in the *RulePoint Administrator Guide* .

You can configure the application and system services in RulePoint for high availability, and reassign objects across the configured application services for processing. For more information, see the "High Availability" chapter in the *RulePoint Administrator Guide*.

Informatica Real-Time Alert Manager

Informatica Real-Time Alert Manager (RTAM) is a web-based dashboard to receive alerts from RulePoint.

You can group the RTAM alerts as channels. Each alert has a priority, subject, and body. RulePoint includes a standard RTAM responder to send alerts to RTAM.

The Proactive Monitoring solution provides two standard modes of alert delivery, through email and RTAM. You can configure the solution to get alerts through email, RTAM, or both. In case of RTAM, you can log in to the RTAM web application and see the alerts on-demand.

Proactive Monitoring Repository

The Proactive Monitoring repository stores the solution metadata, solution configuration data, and the data collected from the PowerCenter services and hosts.

The repository stores a cache of historical data of workflow and session executions. The cache populates as the solution monitors executions to reduce the number of queries against the PowerCenter repository, and improve the overall throughput of the solution. The repository also stores the history of alerts generated by the solution.

The solution includes a set of analytics that use the Proactive Monitoring repository as part of rule processing to add additional details, such as CPU and memory consumption information to the alerts.

Proactive Monitoring for PowerCenter Management Console

Proactive Monitoring for PowerCenter Management Console is a web-based application for configuring the Proactive Monitoring solution.

The Management Console allows users to add details of PowerCenter hosts, nodes, grids, files, folders, and application services for monitoring purposes. The application services include PowerCenter Repository Service, PowerCenter Integration Service, and Web Services Hub. The Management Console allows user to change the monitoring solution settings and the mode of alert delivery, whether to use email or RTAM.

You can use the Management Console to manage the monitored folder lists, topics, sources, watchlists, and rules. You can deploy, undeploy, or redeploy PMPC SQL source services and rules, edit topics, and view the statistics of a rule activation. Use the Reports dashboard to view the health of the services and to get information on the alerts that occur over a period of time.

Node Agent

The node agent is a Proactive Monitoring for PowerCenter component that runs on each of the PowerCenter nodes and collects resource usage information from the host machine.

The node agent periodically gathers CPU, memory, file and folder statistics, and process lifecycle details from the host machine and stores this information in the Proactive Monitoring repository. The resource consumption information is used in rule processing and to add additional information to the alerts.

Solution Usage

The Proactive Monitoring solution connects to the PowerCenter environment with minimum configuration requirements.

Perform the following types of tasks to monitor PowerCenter operations:

Installation

Install Proactive Monitoring for PowerCenter on a machine that is separate from the host machines in the PowerCenter domain.

The solution objects connect to the PowerCenter repository to gather operational data like completed sessions or completed workflows. To fetch data from the tables and views in the PowerCenter repository, the solution requires creation of a read-only user with specific privileges.

After installation, you must copy and run the node agent component binaries on all nodes where the PowerCenter services run.

Configuration and customization

The solution needs to connect to a specific Informatica domain to start monitoring the domain. This involves configuring the solution through Proactive Monitoring Management Console.

When the solution is online, you can customize and extend built-in rules to enhance the monitoring capabilities based on the business needs.

Receive alerts

The solution can be configured to send email and RTAM alerts to the personas defined in the solution, namely pcadmin, pcmonitor, dataarchitect, apparchitect, itsecurity. Each of these personas can have an associated RTAM login or email ID. In case of email, the user will start receiving email alerts as and when anomalies are detected by the solution. In case of RTAM, the user needs to login to the RTAM web application to check the alerts on demand.

You can also use the PMPC solution to send alerts as SNMP traps in addition to email and RTAM alerts. Proactive Monitoring for PowerCenter supports SNMP v2.

Viewing reports

You can use the Reports dashboard to get information about the health of the monitored services, hosts, and nodes in a Informatica domain. View the execution failures that occur over a period of time and drill down into the alert details from the Reports dashboard.

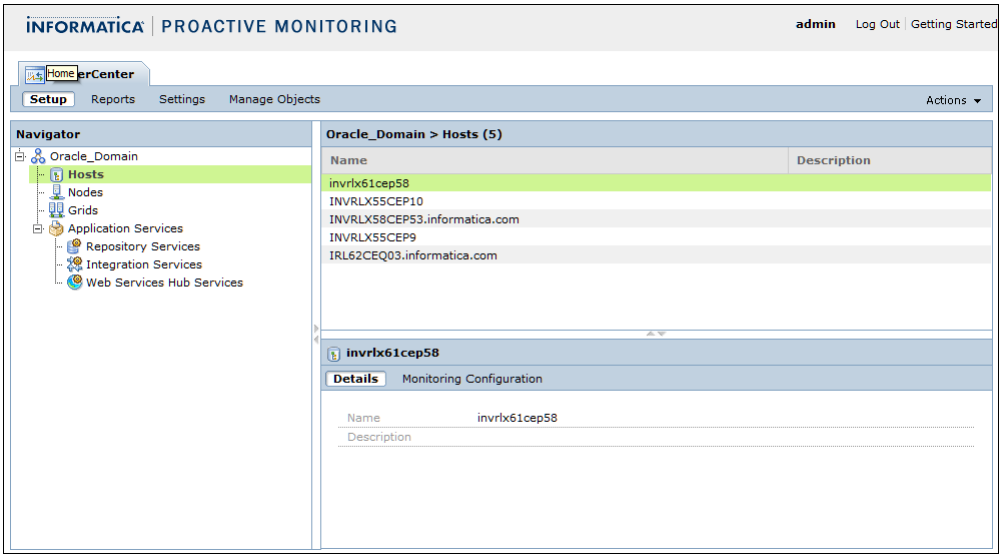
Manage objects

You can use the Manage Objects tab to manage watchlists, SQL sources, rules, and topics. You can view the events generated for a source and the activations for a rule.

Proactive Monitoring Management Console

The Proactive Monitoring Management Console (Management Console) includes the Setup, Reports, Settings, and Mange Objects tab.

The following image shows the Proactive Monitoring Management Console:



Setup Tab

You can use the Setup tab to configure the Informatica domain to monitor. You can provide details about the hosts, nodes, and the application services of the domain in the Setup tab.

Reports Tab

You can use the Reports tab to monitor the Repository Services, Integration Services, and hosts within a single Informatica domain. You can also analyze alerts and drill down into the current and historical alerts.

Settings Tab

You can use the Settings tab to configure the default global settings to receive alerts and the default alert recipients. You can also configure the source timestamp for each workflow from its startup time.

Manage Objects Tab

You can use the Manage Objects tab to manage the list of PowerCenter folders, topics, rules, watchlists, and sources that you use to monitor PowerCenter.

Logging In to the Proactive Monitoring Management Console

Use the Proactive Monitoring for PowerCenter URL to log in to the Management Console. When you log in to the Proactive Monitoring Console, you must specify the user name and password.

1. Open a web browser.
2. In the **Address** field, enter the URL for the Proactive Monitoring Console according to the security within the environment.
   - If you enabled secure connections during installation, enter the following URL:

     `https://<host name>:<port number>/pmpc`
   - If you did not enable secure connections during installation, enter the following URL:

     `http://<host name>:<port number>/pmpc`

   The port number is the port of the tomcat server. Default is 8080.

   The **Informatica Proactive Monitoring** login page appears.

3. Enter the user name and password.
4. Click **Log In**.

CHAPTER 2

Monitoring PowerCenter Operations

This chapter includes the following topics:

- Monitoring PowerCenter Operations Overview, 17
- Proactive Monitoring Objects, 17
- Proactive Monitoring Rules, 19
- Monitoring PowerCenter Processes and Hosts, 21

Monitoring PowerCenter Operations Overview

The Proactive Monitoring solution retrieves information from the PowerCenter environment at regular intervals to provide the benefit of continuous operational monitoring.

The Proactive Monitoring solution retrieves the following information:

- Completed sessions and workflows data
- Running sessions and workflows data
- CPU and memory consumption information from domain nodes
- File and folder statistics

The solution includes various sources, analytics, and responders that connect to PowerCenter services and nodes to collect and process data.

Proactive Monitoring Objects

Proactive monitoring includes a set of pre-defined objects that connect to PowerCenter services in order to drive rule processing and alerting. In the RulePoint programming model, the objects that connect to external systems are broadly classified as sources, analytics, responders, and responses. These objects are configurable and can link to other systems, such as email, Real-Time Alert Manager, or a database.

These predefined objects have the following function:

- Sources gather information from a system.

- Analytics analyze data within a system.
- Responders execute a response through a system.
- Responses are where you define how you want RulePoint to respond if the event matches the rule condition.

Sources

Proactive monitoring source services connect to PowerCenter services, collect data and turn the data into events for rule processing.

The sources execute against the respective PowerCenter services in a pre-defined interval and collect incremental data changes. For example, a query to retrieve completed sessions information from the PowerCenter repository runs every 10 minutes and selects the completed sessions recorded in the 10 minutes by using the time stamp from the previous run.

The solution contains the following types of pre-defined sources:

PMPC SQL Source

The PMPC SQL Source is a custom built SQL source for the Proactive Monitoring solution. The source can connect to multiple repository databases and run SQL queries in parallel. In addition, the PMPC SQL source includes queries relevant for the database types, Oracle, IBM DB2, and Microsoft SQL Server.

All PMPC SQL related services use the repository configuration provided through the Proactive Monitoring for PowerCenter Management Console to connect to the PowerCenter repository databases.

The solution includes multiple instances of PMPC SQL Source that run predefined SQL queries against the configured PowerCenter repository databases at regular intervals.

The PMPC SQL Sources run the SQL queries to create events and publish these events on the following predefined topics:

pc_completed_sessions

pc_completed_workflows

pc_concurrent_workflows

pc_running_sessions

pc_running_sessions_count

pc_running_workflows

pc_missed_workflows

pc_scheduled_workflows

pc_session_to_workflow_ratio

pc_sessions

For example, *PowerCenter Completed Sessions* is the PMPC SQL Source that connects to the PowerCenter Repository, retrieves information about sessions that have completed in the last 10 minutes, and publishes them as events on the pc_completed_sessions topic.

PMPC WSH Ping Service

The PMPC WSH Ping Service connects to PowerCenter Web Services Hub instances to check for the availability of repository services and their associated integration services.

All PMPC WSH related services use the Web Services Hub configurations provided in the Proactive Monitoring for PowerCenter Management Console to connect to the web service hub instances.

Data collected by this source is converted into events and published on the predefined topic, pc_ping.

PMPC WSH Runtime Statistics Receiver

The PMPC WSH Runtime Statistics Receiver connects to PowerCenter Web Services Hub instances to get information on running workflows and sessions at any given point in time.

Data collected by this source is converted into events and published on the predefined topic, pc_runtime_stats.

The solution also includes instances of other predefined RulePoint source types to manage solution specific internal data, such as cache and purge management.

Analytics

The solution contains pre-defined SQL analytics that are used in rules.

As part of rule processing, the SQL analytics run pre-defined queries against the proactive monitoring repository database or the configured PowerCenter repositories on demand.

For examples, the SQL Analytic, get_no_session , gets the total number of sessions from the PowerCenter repository and the SQL Analytic, pc_get_email, gets the email address for a specified alert recipient from the proactive monitoring repository

The solution also uses analytics to fetch CPU and memory consumption details from the hosts that you configured for monitoring and adds these details into the body of the email and RTAM alert notifications.

Responders

The solution contains pre-defined responders that dispatch alerts to external systems.

The proactive monitoring solution includes the following responders:

- The email responder sends email alerts to the various personas.
- The RTAM responder sends RTAM alerts to the various personas.
- The PMPC WSH Workflow Controller restarts workflows in response to the detection of unresponsive workflows. The responder connects to Web Services Hub service to either restart or abort workflows. The responder receives the details of the PowerCenter Repository Service and PowerCenter Integration Services as parameters so that it can detect an appropriate Web Services Hub from the configured set of Web Service Hubs.
- The SNMPv2 responder sends SNMP traps to the network manager that you configure to receive traps.

Proactive Monitoring Rules

The Proactive Monitoring solution contains a set of predefined rules and rule templates that detect anomalies within the PowerCenter environment.

Types of Rules

The predefined rules are categorized based on the types of checks that they perform.

You can use the following types of rules to monitor PowerCenter operations:

Service failures

The solution continuously monitors the Informatica domain, repository databases, and the application services. Application services include the PowerCenter Integration Service, PowerCenter Repository

Service, and Web Services Hub. The solution sends an alert when any service goes down. Immediate notification of such failures allows the administrators to respond to service disruptions quickly.

Resource usage on host

PowerCenter run-time can consist of a grid of hosts and database systems whose CPU, memory, and tablespace availability impacts workflow and session processing. The solution continuously monitors hosts for resource consumption patterns and sends an alert to the administrator if the consumption exceeds user specified thresholds. Timely notification of resource usage allows administrators to proactively manage resource allocations. The solution can also monitor files and folders on a host.

Execution failures

Production deployments of PowerCenter may involve large number of session executions every day. Detailed analysis of session completion data is critical for the overall data movement process. The solution continuously monitors completed workflow and session executions to check for counts of failed rows, zero rows, and non-responsive executions and alerts PowerCenter data architects. The alerts help detect downstream data processing errors quickly and effectively.

Service Level Agreement (SLA) violations

PowerCenter workflow and session executions may need to meet strict execution time SLAs to ensure timely delivery of data to downstream applications and business processes. The solution continuously analyzes workflow or session completion times, and compares them against user defined SLAs to identify any violations. SLA violation alerts enable administrators and architects to analyze the workflow or session definitions, and tune them for better performance.

Execution deviations

An execution deviation is a special case of SLA violations where the solution compares session completion times against historical completion times to determine deviations. Users can specify the minimum deviation beyond which the solution generates an alert when the session or workflow completion time exceeds the historical completion time.

Execution failures due to design changes

Developer changes to PowerCenter objects may cause execution failures. The solution monitors both metadata changes and execution failures, and can correlate the change to the failure. Alerts that originate from these checks contain information on the modified object , user who performed the change, and the time of execution. Detailed alert allows developers and architects to detect the cause of execution failures and take corrective action effectively.

Missed schedules

Schedule misses may cause a delay and SLA violations for downstream applications and business processes. The solution monitors workflow executions and compares with specified schedules to detect deviations and alert the PowerCenter users.

Templates

The Proactive Monitoring solution provides predefined rule templates that include built-in checks and customizable parameters.

Templates enable users to leverage the base logic and customize it for many use cases. For example, a template that checks for SLA violation includes all the predefined conditions to detect the SLA violation. The template also allows the user to specify the workflow or the percentage of deviation that triggers an alert.

You can create template rules from templates. The solution includes a set of template rules to illustrate their usage and behavior.

The template, *PC_OWT7 Workflow running time exceeds the folder SLA*, has built-in checks to detect if the workflow running time exceeds the specified folder SLA. You can customize the folder SLA. The template

rule, *PC_OW2 Workflow running time exceeds folder SLA of 5 minutes*, is created using this template with the folder SLA parameter set to 5 minutes.

Similarly, you can create more rules with this template for folders with different SLAs.

Advanced Rules

Advanced rules do not provide any parameterization.

For example, a rule that checks for zero loaded rows may not require any parameterization. The Proactive Monitoring solution provides many such advanced rules.

Monitoring PowerCenter Processes and Hosts

The Proactive Monitoring solution monitors the health of PowerCenter processes, along with the CPU and memory resource consumption on the host machines where they are run.

The alert includes CPU and memory consumption information of the host during the execution time of workflows and sessions. You can use the information to understand which processes were running on the host that consumed memory and CPU when the workflow and sessions were running. For example, the session might exceed the Service Level Agreement because one of the processes consumed CPU at that time and hence the delay. The information will help you to troubleshoot issues, perform effective root cause analysis, and take corrective actions.

Collection of CPU, Memory, and Process Health Information

The Proactive Monitoring solution provides the Node Agent component to collect resource utilization and lifecycle data of processes that run on a host.

The node agent continuously collects information about CPU and memory utilization of processes that run on a host and records the information in the Proactive Monitoring repository. The CPU usage value is a calculation of the total CPU usage divided by the number of cores. Memory usage is a percentage of the memory used by the process divided by the total memory available.

Configure the node agent through the Proactive Monitoring Management Console. You can specify the frequency for data collection, how many processes to collect data on, and the processes to monitor for lifecycle changes. The node agent connects to the Proactive Monitoring repository database through a JDBC connection. The node agent then collects statistics from the host and stores the details in the Proactive Monitoring repository database. It can monitor the lifecycle of a process, such as pmdtm, and generate an alert when the process stops.

The source, *Node TopNStats Table Purge*, purges the statistics from the Proactive Monitoring repository database at a scheduled interval of two hours. Start the *Node TopNStats Table Purge* source with the other Proactive Monitoring sources. If the node agent on a monitored node fails to respond with the statistics details from the node, the Proactive Monitoring solution triggers the *PC_O16 Node Agent health check* rule. This rule alerts the pcadmin persona that the node agent running on the particular node is down.

Use the Reports dashboard to get information about the health of the monitored services, hosts, and nodes in the Informatica domain.

Node Agent Statistics

Node agent collects statistics from each monitored node and stores the information in the Proactive Monitoring repository database.

The following table shows the statistics collected by the node agent from each monitored node:

Property	Description
System CPU utilization	Total CPU usage of the monitored node.
System memory utilization	Total memory usage of the monitored node.
Total number of processes	Total number of processes running on the monitored node.
PID	A number that identifies a process while it runs. The node agent collects the PID of the top processes as configured in the Proactive Monitoring for PowerCenter Management Console.
CPU usage	The percentage of CPU used by a process. The node agent collects the CPU usage of the top processes as configured in the Proactive Monitoring for PowerCenter Management Console.
Memory usage	The percentage of memory used by a process. The node agent collects the memory usage of the top processes as configured in the Proactive Monitoring for PowerCenter Management Console.
Name	Name of processes to be monitored.
Arguments	Command arguments of the processes that you want to monitor.
Elapse Time	Time spent by a process on the node since it was started. The node agent collects the elapse time statistics for the top processes.

CHAPTER 3

Manage Objects

This chapter includes the following topics:

- Manage Objects Overview, 23
- Managing PMPC SQL Source Services, 25
- Monitored Event Management, 26
- Monitored Objects Management, 26
- Managing Templates Rules, 31

Manage Objects Overview

You can manage the monitored topics, watchlists, sources, and rules from the **Manage Objects** tab.

Perform the following tasks from the **Manage Objects** tab:

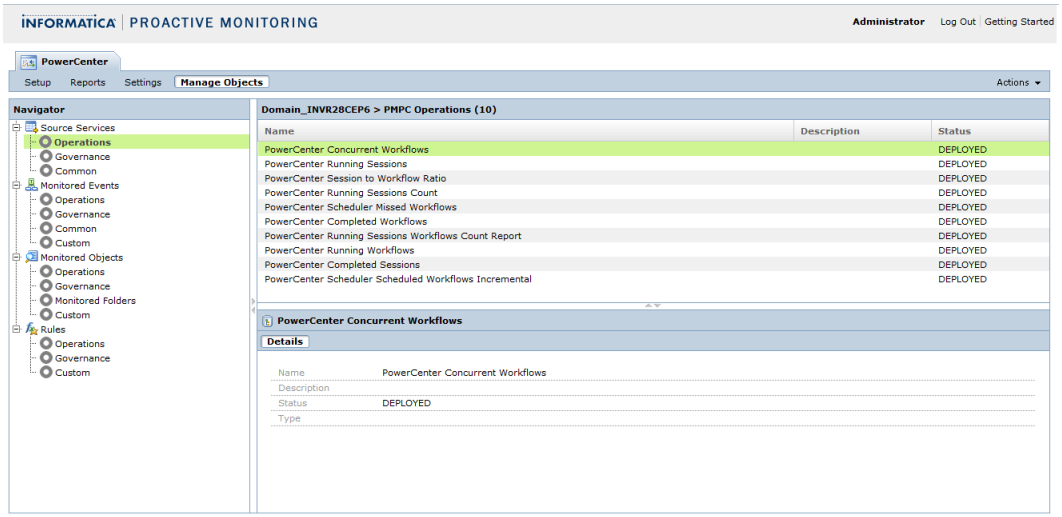
- Manage the PowerCenter folders that you monitor from the PowerCenter Monitored Folders watchlist. You can add a folder that you want to monitor to the PowerCenter Monitored Folders watchlist or remove a folder that you do not want to monitor from the PowerCenter Monitored Folders watchlist.
- Deploy, undeploy, or redeploy the PMPC SQL source services.
- View the statistics of events generated for a topic, including the event count and the deployment details.
- View and edit watchlists
- View the activation count, status, and details of a rule. You can also view the statistics along with the number of activations for a selected timeline.
- Create, edit, copy, deploy, undeploy, redeploy, edit, or delete template rules.

For more information on the state of objects and deployment related actions, see the "Deployment Overview" chapter in the *RulePoint Administrator Guide*.

Manage Objects View

In the **Manage Objects** tab, you can manage the source services, topics, watchlists, and rules.

The following figure shows the **Manage Objects** tab:



The **Manage Objects** tab has the following components:

Navigator

Appears in the left pane of the **Manage Objects** tab. The Navigator displays the following entities that you can monitor from the Proactive Monitoring for PowerCenter Management Console:

- PMPC source services. Deploy or undeploy source services from the Proactive Monitoring for PowerCenter Management Console.
- Monitored events. View the statistics of generated events.
- Monitored objects. View the monitored folders where you can manage the list of PowerCenter folders to monitor. You can also manage watchlists.
- PMPC rules. Manage template rules.

Contents panel

Appears in the right pane of the **Manage Objects** tab and displays information about monitored events, objects, rules, folders, and source service that you select in the Navigator.

Actions menu

When you select a source service in the contents panel, you can deploy, undeploy, or redeploy a source service.

When you select monitored events in the contents panel, you can view the event count and status of the topic. You can also view the events arrived for a the associated source for a selected timeline.

When you select monitored objects in the contents panel, you can edit that object. You can also deploy, undeploy, or redeploy the object. When you select monitored folders, you can add or remove the monitored folders.

When you select a template rule, you can create, edit, copy, delete, deploy, undeploy, or redeploy template rules. You can also view the statistics of a rule activation, including the status of the rule and the rule details.

Managing PMPC SQL Source Services

You can use the PMPC SQL source service to connect to a database and run SQL queries or commands to create RulePoint events. You can deploy, undeploy, or redeploy PMPC SQL source services from the Proactive Monitoring for PowerCenter Management Console.

The PMPC SQL source is a custom built SQL source for the proactive monitoring solution. You can manage PMPC SQL source services from the **Manage Objects** tab in the Proactive Monitoring for PowerCenter Management Console.

Note: To manage other SQL source services, use RulePoint.

Deploying, Undeploying, or Redeploying SQL Source Services

Use the **Manage Objects** tab in the Proactive Monitoring for PowerCenter Management Console to deploy, undeploy, or redeploy Proactive Monitoring for PowerCenter source services.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Manage Objects** tab.
2. In the Navigator, select the **Operations**, **Governance**, or **Common** source service.

   The **Operations** source service lists the predefined source services that are available by default after you install Proactive Monitoring for PowerCenter Operations. The **Governance** source service lists the predefined source services that are available by default after you install Proactive Monitoring for PowerCenter Governance. The **Common** source service lists the predefined source services that are common to both Operations and Governance.

   Based on the source service that you select in the Navigator, the list of predefined source services along with the description and status information appear in the contents panel. You can view the details of the source service from the **Details** view in the contents panel.
3. To deploy a source service, perform the following tasks:
   a. Select the source service, and click **Deploy** on the **Actions** menu.

      You can deploy an object that is in the Drafts state.

      A message appears that indicates that the source is deployed successfully.
   b. Click **OK**.

      The status of the source service changes from Draft to Deployed.
4. To undeploy a source service, perform the following tasks:
   a. Select the source service, and click **Undeploy** on the **Actions** menu.

      You can undeploy a source that is in the Deployed or Needs_Deployment state.

      A message appears that indicates that the source is undeployed successfully.
   b. Click **OK**.

      The status of the source service changes from Deployed to Draft.
5. To redeploy a source service, perform the following tasks:
   a. Select the source service, and click **Redeploy** on the **Actions** menu.

      You can redeploy a source when you edit a deployed source. The state of the source changes from Deployed to Needs_Deployment.

      A message appears that indicates that the source is redeployed successfully.
   b. Click **OK**.

      The status of the source service changes from Needs_Deployed to Deployed.

For more information about object states and deployment, see *RulePoint Administrator Guide*.

Monitored Event Management

The Monitored Events view contains the list of monitored topics for PowerCenter and the events generated for a source. You can view the monitored Proactive Monitoring for PowerCenter topics from the **Manage Objects** tab in the Proactive Monitoring for PowerCenter Management Console.

You can view the statistics only when the topics and the associated objects are deployed. You might not be able to view the event statistics for a source if you have deployed only the topic and not the source associated with the topic. You also cannot view statistics for topics that do not have an associated source and are system generated.

Viewing Topics

View all the monitored topics, the events generated for each of the objects, and the status of the topics in the Management Console.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Manage Objects** tab.
2. In the Navigator, select the **Operations**, **Governance**, **Common**, or **Custom** topics.

Option	Description
Operations	Lists the predefined topics that are available by default after you install Proactive Monitoring for PowerCenter Operations.
Common	Lists the predefined topics that are common to both Operations and Governance.
Custom	Lists the topics that you create in RulePoint.

   Based on the topic that you select in the Navigator, the list of topics along with the description and status information display in the contents panel. Proactive Monitoring for PowerCenter takes some time to fetch the data and display the list of monitored events. You can view the details of the topic you select from the **Details** view in the contents panel.
3. To view the statistics of the events generated for a source associated with the selected topic, perform the following steps:
   a. Select the topic, and then select **Stats** from the **Actions** menu.
   b. Select a timeline to view the events generated for that period.

   The graph lists the statistics of the events, including the number of events generated for the source. Hover the mouse over the graph to view the activation for a particular period.
4. Click **Cancel** to exit from the view.

Monitored Objects Management

You can view and manage the watchlists and the monitored folders from the **Manage Objects** tab in the Proactive Monitoring for PowerCenter Management Console.

Viewing and Editing Watchlists

Use the **Manage Objects** tab in the Proactive Monitoring for PowerCenter Management Console to view or edit the watchlists. When you edit a watchlist and save it, that watchlist along with the referenced primary objects is also deployed.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Manage Objects** tab.
2. In the Navigator, select the **Operations**, **Governance**, **Common**, or **Custom** watchlist.

   The **Operations** watchlist lists the predefined watchlists that are available by default after you install Proactive Monitoring for PowerCenter Operations. The **Governance** watchlist lists the predefined topics that are available by default after you install Proactive Monitoring for PowerCenter Governance. The **Common** watchlist lists the predefined watchlists that are common to both Operations and Governance. The **Custom** watchlist lists the watchlists that you create.

   Based on the watchlist that you select in the Navigator, the list of predefined watchlists along with the name and status information appear in the contents panel. The details panel displays the details of the selected watchlist.
3. To edit a watchlist, select the watchlist, and then select **Edit** from the **Actions** menu.
4. In the **Content** field, edit the existing values or add additional values in a new line for the list **Type**.
5. Click **Save**.

   The watchlist along with the supporting objects is also deployed. The state of the object changes to Deployed.

Monitored Folder Management

PowerCenter Monitored Folders is a predefined watchlist that contains the list of PowerCenter folders to monitor. You can manage the monitored folders from the **Manage Objects** tab in the Proactive Monitoring for PowerCenter Management Console.

You manage the list of PowerCenter folders that you monitor by adding or removing folders from the list.

To receive notifications add your folder names to this watchlist. If the folder is same for multiple PowerCenter repositories, you receive notification for all configured repositories.

Adding or Removing Folders from Monitored Folders

You can add or remove folders to the predefined list of PowerCenter folders that you monitor.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Manage Objects** tab.
2. In the Navigator, select **Monitored Objects** and then click **Monitored Folders**.

   The list of folders that you monitor appears in the contents panel.
3. On the **Actions** menu, click **Add/Remove**.

   The **Add/Remove Monitored Folders** screen appears.
4. To add a folder or folders to the list of monitored folders, select and move the folder from the **All Folders** section to the **Monitored Folders** section.
   a. Select a Repository Service to display the folders in the repository. You can enter a folder name and use the **Filter** button to filter the folders based on the folder name.
   b. Select the folder that you want to add to the list of monitored folders. You can select multiple folders at the same time. To select all the folders of a repository service, double-click on the repository service.

c. To add the folder that you select to the list of monitored folders, click the **>>** button.

The folder name appears in the **Monitored Folders** section.

5. To remove a folder or folders from the list of monitored folders, select and move the folder from the **Monitored Folders** section to the **All Folders** section.
   a. Select a Repository Service to display the folders in the repository that you monitor. You can enter a folder name and use the **Filter** button to filter the folders based on the folder name.
   b. Select the folder that you want to remove from the list of monitored folders. You can select multiple folders at the same time. To select all the folders of a repository service, double-click on the repository service.
   c. To remove the folder from the list of monitored folders, click the **<<** button.

      The folder name appears in the All Folders section.
6. Click **Save** to save the changes that you made to the monitored folders list.

   A message appears that the Monitored Folders are updated successfully. You need to update the runtime with the changes.
7. To update the runtime with the changes, select **Update Runtime** from the **Actions** menu.

   A message appears that indicates that the runtime update will take a few minutes.
8. Click **OK**.

   A message appears that the runtime is successfully updated.

Monitor Files and Directories on a Node

The following scenarios explain how you can set the criteria in Proactive Monitoring for PowerCenter to monitor a file or folder in PowerCenter and alert you appropriately.

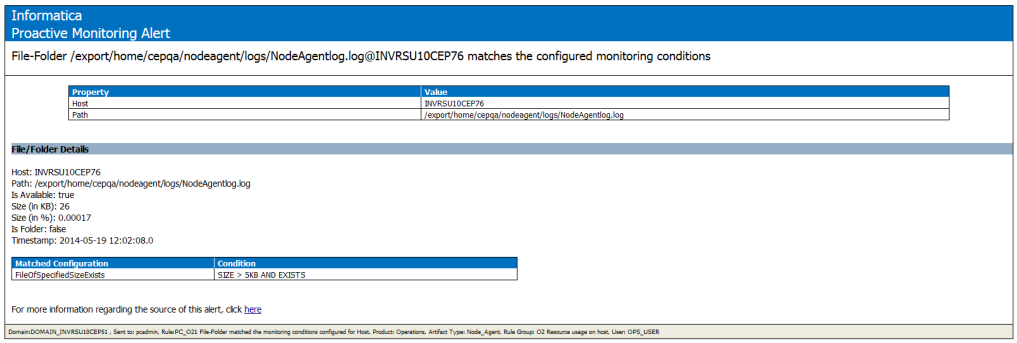
Example - Monitor File Size

You want to monitor if a file of a specified size exists on the PowerCenter host. For example, you want to monitor the workflow logs folder, `INFA_HOME/server/infa_shared/WorkflowLogs`. You need to monitor if a specific file, `wf_pmpc_reports.log.RUNINSTANCE3.5230.20140514121201.bin`, exists within this folder. You also want to check if the log file exceeds the specified threshold limit.

1. Select the host whose file you want to monitor, and provide the criteria in the **Add Monitored Path** page.
2. Provide a valid configuration name for the monitoring criteria.
3. Enter the path of the file that you want to monitor.
4. Select the **Path Type** as **File**, and choose the criteria based on which you want to monitor the file:
   a. Select **size > than 5 KB**.
   b. Select **Path Exits**.

5. Click **Save**.

   The following image shows the alert in RTAM when the rule activates:


Informatica
Proactive Monitoring Alert

File-Folder /export/home/cepqa/nodeagent/logs/NodeAgentlog.log@INVRSU10CEP76 matches the configured monitoring conditions

Property	Value
Host	INVRSU10CEP76
Path	/export/home/cepqa/nodeagent/logs/NodeAgentlog.log

File/Folder Details

Host: INVRSU10CEP76
Path: /export/home/cepqa/nodeagent/logs/NodeAgentlog.log
Is Available: true
Size (in KB): 26
Size (in %): 0.00017
Is Folder: false
Timestamp: 2014-05-19 12:02:08.0

Matched Configuration	Condition
FileOfSpecified5izeExists	SIZE > 5KB AND EXISTS

For more information regarding the source of this alert, click here

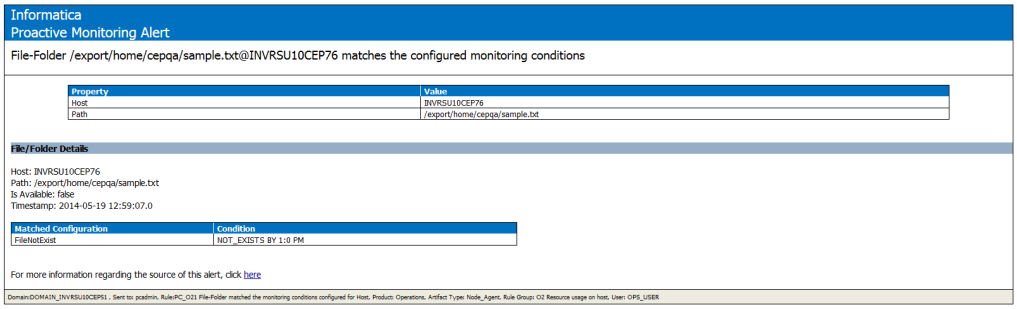
Domain:DOMAIN_INVRSU10CEPS1 , Sent to: pcadmin, Rule:PC_O21 File-Folder matched the monitoring conditions configured for Host. Product: Operations. Artifact Type: Node_Agent. Rule Group: O2 Resource usage on host. User: OPS_USER

Example - Monitor Removal of a File

You run a workflow that involves a command task to delete a specific file. You want an alert when the file is removed in the host by a specified time.

1. Select the host whose file you want to monitor, and provide the criteria in the **Add Monitored Path** page.
2. Provide a valid configuration name for the monitoring criteria.
3. Enter the path of the file that you want to monitor.
4. Select the **Path Type** as **File**, and choose the criteria **Path does not exist**.
5. Select **Time Filter**, and then select the condition **BY 1:00 PM**.
6. Click **Save**.

   The following image shows the alert in RTAM when the rule activates:


Informatica
Proactive Monitoring Alert

File-Folder /export/home/cepqa/sample.txt@INVRSU10CEP76 matches the configured monitoring conditions

Property	Value
Host	INVRSU10CEP76
Path	/export/home/cepqa/sample.txt

File/Folder Details

Host: INVRSU10CEP76
Path: /export/home/cepqa/sample.txt
Is Available: false
Timestamp: 2014-05-19 12:59:07.0

Matched Configuration	Condition
FileNotExist	NOT_EXISTS BY 1:0 PM

For more information regarding the source of this alert, click here

Domain:DOMAIN_INVRSU10CEPS1 , Sent to: pcadmin, Rule:PC_O21 File-Folder matched the monitoring conditions configured for Host. Product: Operations. Artifact Type: Node_Agent. Rule Group: O2 Resource usage on host. User: OPS_USER

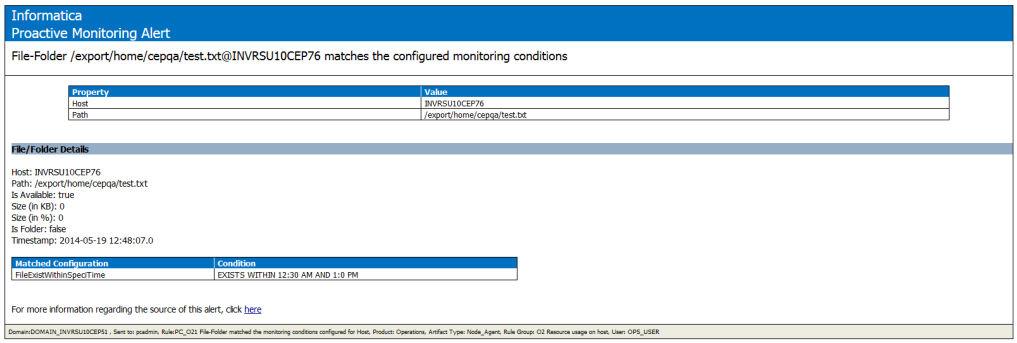
Example - Monitor Creation of a File

You want to monitor if a file exists at a specified time on the PowerCenter host. When you expect to run a workflow at a specified time interval, you want an alert after the log file for that workflow execution is created.

1. Select the host whose file you want to monitor, and provide the criteria in the **Add Monitored Path** page.
2. Provide a valid configuration name for the monitoring criteria.
3. Enter the path of the file that you want to monitor.
4. Select the **Path Type** as **File**, and select the criteria **Path exists**.
5. Select **Time Filter**, and then select the condition **WITHIN 12:30 PM AND 1:00 PM**.

6. Click **Save**.

   The following image shows the alert in RTAM when the rule activates:


Informatica
Proactive Monitoring Alert

File-Folder /export/home/cepqa/test.txt@INVRSU10CEP76 matches the configured monitoring conditions

Property	Value
Host	INVRSU10CEP76
Path	/export/home/cepqa/test.txt

File/Folder Details

Host: INVRSU10CEP76
Path: /export/home/cepqa/test.txt
Is Available: true
Size (in KB): 0
Size (in %): 0
Is Folder: false
Timestamp: 2014-05-19 12:48:07.0

Matched Configuration	Condition
FileExistWithinSpecTime	EXISTS WITHIN 12:30 AM AND 1:0 PM

For more information regarding the source of this alert, click here

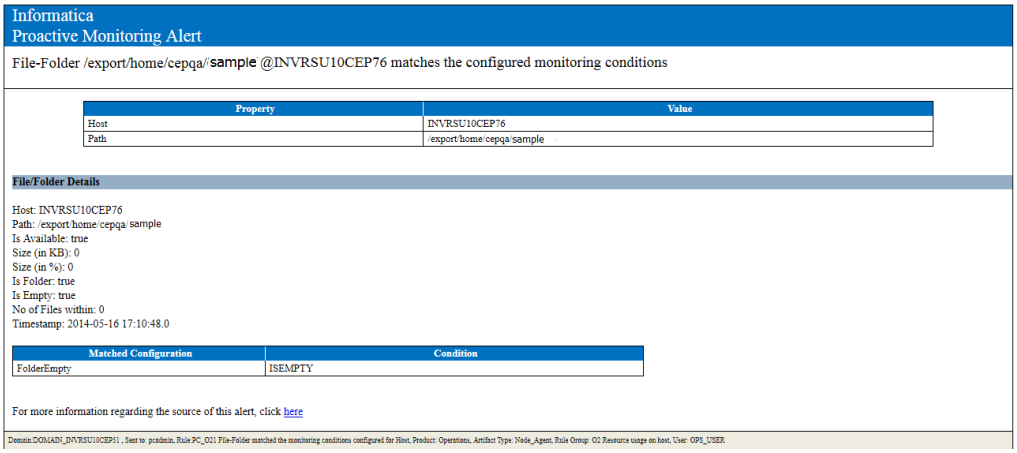
Domain:DOMAIN_INVRSU10CEP51 , Sent to: pcadmin, Rule:PC_O21 File-Folder matched the monitoring conditions configured for Host, Product: Operations, Artifact Type: Node_Agent, Rule Group: O2 Resource usage on host, User: OPS_USER

Example - Monitor Empty Paths

You want to verify that the workflow logs directory does not have content.

1. Select the host whose file you want to monitor, and provide the criteria in the **Add Monitored Path** page.
2. Provide a valid configuration name for the monitoring criteria.
3. Enter the path of the folder that you want to monitor.
4. Select the **Path Type** as **Folder**, and choose the criteria **Is Path Empty**.
5. Click **Save**.

   The following image shows the alert in RTAM when the rule activates:


Informatica
Proactive Monitoring Alert

File-Folder /export/home/cepqa/sample @INVRSU10CEP76 matches the configured monitoring conditions

Property	Value
Host	INVRSU10CEP76
Path	/export/home/cepqa/sample

File/Folder Details

Host: INVRSU10CEP76
Path: /export/home/cepqa/sample
Is Available: true
Size (in KB): 0
Size (in %): 0
Is Folder: true
Is Empty: true
No of Files within: 0
Timestamp: 2014-05-16 17:10:48.0

Matched Configuration	Condition
FolderEmpty	ISEMPTY

For more information regarding the source of this alert, click here

Domain:DOMAIN_INVRSU10CEP51 , Sent to: pcadmin, Rule:PC_O21 File-Folder matched the monitoring conditions configured for Host, Product: Operations, Artifact Type: Node_Agent, Rule Group: O2 Resource usage on host, User: OPS_USER

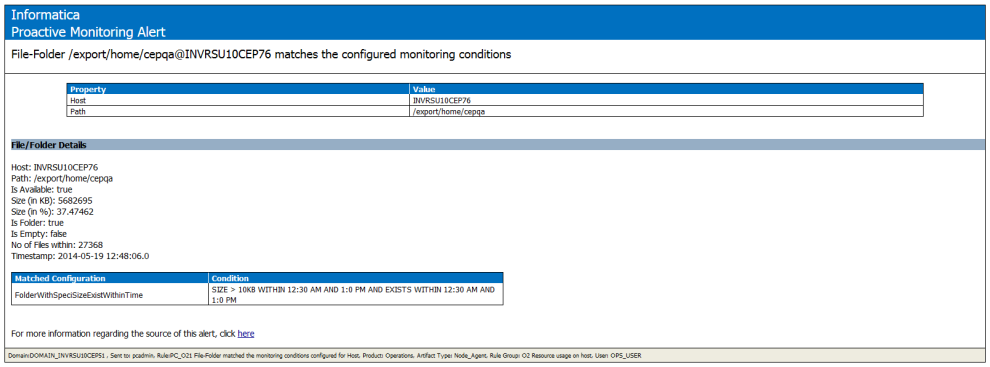
Example - Monitor Directory Size Within Time Period

You want to monitor the size of the workflow logs folder and receive an alert when the folder size exceeds a specified threshold within a specific time interval. You might want to do this because you want to trigger a script to archive the logs.

1. Select the host whose file you want to monitor, and provide the criteria in the **Add Monitored Path** page.
2. Provide a valid configuration name for the monitoring criteria.
3. Enter the path of the file that you want to monitor.

4. Select the **Path Type** as **Folder**, and select the following criteria:
   a. Select **Size > than 10 KB**.
   b. Select **Path exists**.
5. Select **Time Filter**, and then select the condition **WITHIN 12:30 AM AND 1:00 PM**.
6. Click **Save**.

   The following image shows the alert in RTAM when the rule activates:



Managing Templates Rules

You can view and manage the Proactive Monitoring for PowerCenter template rules from the **Manage Objects** tab in the Proactive Monitoring for PowerCenter Management Console.

The predefined template rules are available by default after you install Proactive Monitoring for PowerCenter Operations or Governance. You can also view the rules that you create other than the available predefined rules under **Custom Monitored Objects**. The contents panel lists the template rules, the activation count, and the status of the monitored rule. You can view the details of the template rule from the **Details** view in the contents panel.

Creating a Template Rule

Use the **Manage Objects** tab in the Proactive Monitoring for PowerCenter Management Console to create a template rule.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Manage Objects** tab.
2. In the Navigator, select **Operations**, **Governance**, or **Custom**.
3. To create a template, select **New** from the **Actions** menu.

   The Template Rule dialog box appears listing the name of the template, the description, and the rule count.
4. Select the template, and then click **Next**.
5. Enter a name for the template rule.
6. Optionally, enter a description for the template rule.
7. In the **Template Parameters** section, provide the properties.
8. Click **Save**.

Editing a Template Rule

When you edit and save a template rule, the template rule is deployed along with the supporting objects that are referenced in the template rule.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Manage Objects** tab.
2. In the Navigator, select **Operations**, **Governance**, or **Custom**.
3. To edit a template, select **Edit** from the **Actions** menu.
4. Edit the template parameters.
5. Click **Save**.

Copying a Template Rule

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Manage Objects** tab.
2. In the Navigator, select **Operations**, **Governance**, or **Custom**.
3. To copy a template rule, select the rule, and then select **Copy** from the **Actions** menu.
4. Enter the name of the template you want to create as a copy.
5. Click **Save**.

Deleting a Template Rule

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Manage Objects** tab.
2. In the Navigator, select **Operations**, **Governance**, or **Custom**.
3. To delete a template, select the template, and then select **Delete** from the **Actions** menu.

   Note: You cannot delete a template rule that is in Deployed state.
4. Click **OK**.

Deploying, Undeploying, and Redeploying Rules

You can deploy, undeploy, or redeploy rules.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Manage Objects** tab.
2. In the Navigator, select **Operations**, **Governance**, or **Custom**.
3. To deploy a template, perform the following tasks:
   a. Select the draft rule that you want to deploy, and then select **Deploy** from the **Actions** menu.

      A message appears that indicates successful deployment. To view the details of the message, click **Details**.
   b. Click **OK**.

      The status of the rule changes from Draft to Deployed.
4. To undeploy a template, perform the following tasks:
   a. Select the deployed rule that you want to undeploy, and then select **Undeploy** from the **Actions** menu.

      A message appears that indicates successful undeployment. To view the details of the message, click **Details**.

b. Click **OK**.

The status of the rule changes from Deployed to Draft.

5. To redeploy a template, perform the following tasks:

a. Select the rule that you want to undeploy, and then select **Redeploy** from the **Actions** menu.

Note: The rule must be in the Needs_Deployment state if you want to redeploy the rule. The rule is in Needs_Deployment when you edit a rule that is in Deployed status.

A message appears that indicates successful redeployment. To view the details of the message, click **Details**.

b. Click **OK**.

The status of the rule changes from Needs_Deployment to Deployed.

For more information on object states and deployment, see *RulePoint Administrator Guide*.

Viewing the Statistics of a Rule Activation

You can view the statistics for a deployed rule. Set a timeline to view the rule activations for that period.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Manage Objects** tab.
2. In the Navigator, select **Operations**, **Governance**, or **Custom**.
3. Select the template rule, and then select **Stats** from the **Actions** menu.
4. Select the timeline to view the statistics of a rule activation for the set timeline.

You can select a timeline between 5 minutes and 24 hours.

The graph lists the statistics of the rule activation, including the number of activations generated for the rule. Hover the mouse over the graph to view the activation for a particular period.

CHAPTER 4

Proactive Monitoring Reports

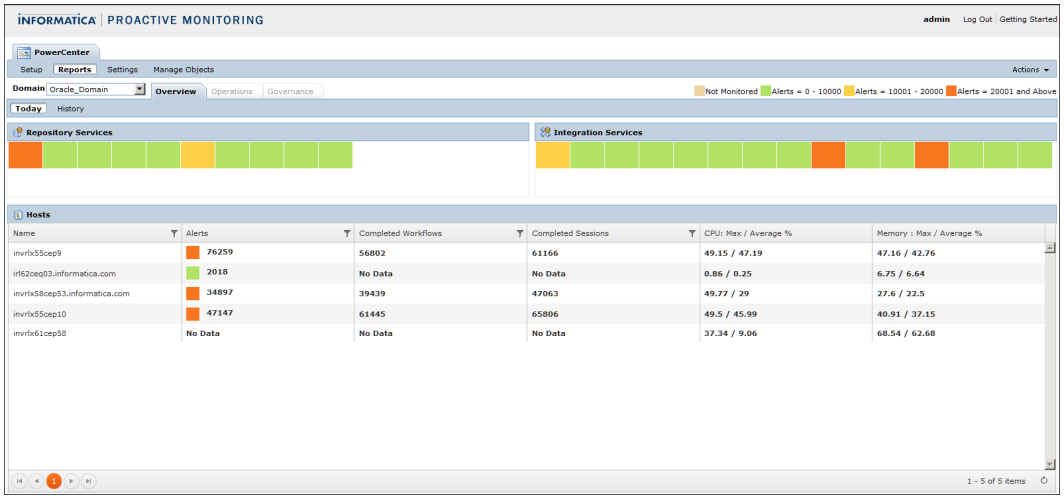
This chapter includes the following topics:

- Proactive Monitoring Reports Overview, 34
- Monitoring Application Services and Hosts in the Domain, 35
- Monitoring PowerCenter Operation Alerts, 37
- Monitoring PowerCenter Governance Alerts, 39

Proactive Monitoring Reports Overview

Use the Proactive Monitoring reports to monitor the Repository Services, Integration Services, and hosts within a single Informatica domain. You can use the Proactive Monitoring reports to analyze and view the current as well as past alerts. You can correlate resource usage on hosts and troubleshoot.

The following image shows the Reports tab in the Proactive Monitoring for PowerCenter Management Console:



You can configure the Reports dashboard to monitor the status of the application services and hosts in a domain. View the best practice violations, execution failures, and node agent alerts that occur over a period of time from the Reports dashboard. Use the on-demand reports to generate reports for workflow, session, or transformation attributes.

You can use the following tabs available in the **Reports** tab to monitor the PowerCenter alerts .

Overview

Monitor the alerts for Repository Services, Integration Services, and hosts in a domain from the **Overview** tab.

Operations

Monitor the PowerCenter execution failures and node agent alerts from the **Operations** tab. You can view the total number of alerts, alerts for a time period, and details of the execution failures and node agent alerts from the **Operations** tab. You can also export the alerts for the current day or for a specified period to a .csv file or an .xlsx file.

Governance

Monitor the violations in the PowerCenter development environment from the **Governance** tab. Generate on-demand reports for workflows, sessions, or transformations. You can view the total number of alerts and alerts for a time period from the **Governance** tab. ou can also export the alerts for the current day or for a specified period to a .csv file or an .xlsx file.

Reports Tab - Filter and Display Options

You can use the filter option to filter the details of the host statistics, execution failures, and best practice violations.

You can filter the values that appear in the contents panel of the **Reports** tab. To filter the details that appear for a column in the **Today** view or **History** view:

1. Click on the filter icon next to the column. The filter conditions appear based on the alert type, application service, or host.
2. Select the filter condition and enter the filter value.
3. Click the **Filter** button to filter the values based on the filter condition and value for that column. The filter icon on the column on which filter condition has been applied appears in orange color.
4. To clear the filter condition for a column, select the filter icon for that column and click the **Clear** button. You can also select "All Alerts" in the alerts table that appears in the **Total Alerts** section to display all the alerts and clear the filter condition.

You can use the **Maximize** icon to scroll to the alert details. To scroll to the alert details and hide the **Total Alerts** section in the **History** view, click the **Maximize** icon. To toggle back to the **Total Alerts** section, click the **Maximize** icon again.

You can also resize the width of the columns that appear in the contents panel to view the complete descriptions.

Monitoring Application Services and Hosts in the Domain

Monitor the status of the Repository Services, Integration Services, and hosts within a single Informatica domain from the **Overview** tab on the **Reports** tab. You can view the CPU, memory usage, running workflows, and running sessions statistics for all the hosts in the domain.

The contents that appear on the **Reports** tab vary based on the view that you select.

You can select the following views:

- **Today**. View the application services alerts and host statistics for the current day.
- **History**. View the application services alerts and host statistics for a particular time period.

In the Reports dashboard, the Repository Services and Integration Services appear in colors that indicate the number of alerts for each service. For example, if the number of alerts for a repository service is less that 50, then the repository service appears in green color. You can customize the alerts range for the colors in which the services appear.

Note: If application services run on multiple hosts for high availability or grid configurations, the alerts appear for the services on all the hosts and not just for the active hosts. In such cases, alert data appears even if application services are not running on the active hosts.

Monitoring Alerts for Application Services and Host Statistics for the Current Day

You can monitor the alerts for the application services and host statistics in the domain for the current day in the **Today** view on the **Overview** tab.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Reports** tab.
2. Click the **Overview** tab and then click the **Today** view. The following sections appear in the Today view:
   Repository Services

   The Repository Services running in the domain appear in brown, green, yellow, or amber color in the **Repository Services** section based on the number of alerts for the Repository Services.

   Integration Services

   The Integration Services running in the domain appear in brown, green, yellow, or amber color in the **Integration Services** section based on the number of alerts for the Integration Services.

   Hosts

   The host name, alerts, number of running workflows and sessions, CPU and memory statistics appear in the **Hosts** section.

   Note: The CPU usage value is the total CPU usage divided by the number of cores. Memory usage is a percentage of the memory used by the process divided by the total memory available.
3. To view the past alerts for a application service, click the **History** link that appears in the tooltip for the application service.

Configuring Display Settings

In the Reports dashboard, the Repository Services and Integration Services appear in colors that indicate the health of the services. The health of the services is based on the number of alerts. You can configure the display settings for alerts in the **Today** view on the **Overview** tab.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Reports** tab.
2. Click the **Overview** tab and then click the **Today** view.
3. In the **Actions** menu on the **Reports** tab, click **Display Settings**.

   The display color of the application services indicate the number of alerts for these services. Application services that you do not monitor appear in brown color.

   The **Settings** screen appears.

4. Specify the upper range of alerts for application services to appear in green and yellow color. For example, if you specify 200 as the upper range of alerts for green color, the application services appear in green color in the **Today** view till the number of alerts for that service exceed 200.
5. Click **Save**.

Monitoring the Alert History for Application Services and Hosts

You can monitor the alert history for a particular time period for PowerCenter application services and hosts in a domain.

You can select multiple Repository Services and Integration Services to monitor, but you can select only one host at a time to monitor.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Reports** tab.
2. Click the **Overview** tab and then click the **History** view.

   The **Total Alerts** and **Alerts for Period** sections appear in the contents panel. A graph that represents the alert history appears in the **Total Alerts** section.
3. Select the time period for which you want display the number of alerts.
   - To select the time period with the slider control, resize the width of slider control to indicate the time period or move the slider control over the time period.
   - To use predefined views to select the time period, click the **1 Day** view, **5 Day** view, **1 Month** view, **3 Months** view, **6 Months** view, or **Custom** view to display the number of alerts for the corresponding period.

   A graph that represents the number of alerts for the application service appears in the **Alerts for Period** section.
4. Select the application service or host for which you want to monitor the alerts.
   - To monitor the alerts for the Repository Service, click the **Repository Services** view and select the Repository Service from the list and then click **Ok**.
     You can select multiple Repository Services to monitor.
   - To monitor the alerts for the Integration Service, click the **Integration Services** view and select the Integration Service from the list and then click **Ok**.
     You can select multiple Integration Services to monitor.
   - To monitor the running sessions, running workflows, CPU statistics, and memory statistics of the host, click the **Host** view and then select the host from the list.

     Note: The CPU usage value is the total CPU usage divided by the number of cores. Memory usage is a percentage of the memory used by the process divided by the total memory available.

Monitoring PowerCenter Operation Alerts

You can monitor the execution failures in PowerCenter and view the alerts for the current day or for a time period.

The contents that appear on the **Operations** tab vary based on the view that you select.

You can select the following views:

- **Today**. View and export the PowerCenter operation alerts for the current day.
- **History**. View and export the PowerCenter operation alerts for a particular time period.

Monitoring the PowerCenter Operation Alerts for the Current Day

You can monitor the PowerCenter operation alerts for the current day in the **Today** view on the **Operations** tab.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Reports** tab.
2. Click the **Operations** tab and then click the **Today** view. The following sections appear in the contents panel:

   Total Alerts

   The alert type, alert count, and percentage of the alert types appear in a graph and table format in the **Total Alerts** section.

   Execution Failures

   The alert details appear in the **Execution Failures** section.
3. To view the execution failures of a particular alert type, click on the corresponding alert type from the donut chart or table. The execution failures for the alert type that you select appear in the **Execution Failures** section.
4. To view detailed information of an alert that appears in the **Execution Failures** section, click the **View Details** button in **Details** column for that alert.

Monitoring the Alert History for PowerCenter Operations

You can monitor the total alerts for PowerCenter operations for a particular time period in the **History** view on the **Operations** tab.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Reports** tab.
2. Click the **Operations** tab and then click the **History** view.

   The **Total Alerts**, **Alerts for Period**, and **Execution Failures** sections appear in the contents panel. A graph that represents the number of alerts over a period of time appears in the **Total Alerts** section.
3. Select the time period for which you want display the number of alerts.
   - To select the time period with the slider control, resize the width of slider control to indicate the time period or move the slider control over the time period.
   - To use pre-defined views to select the time period, click the **1 Day** view, **5 Day** view, **1 Month** view, **3 Months** view, **6 Months** view, or **Custom** view to display the number of alerts for the corresponding period.

   In addition to the pre-defined views, you can use the **Custom** view to create your own time period view.

   For the time period that you select in the **Total Alerts** section, a donut chart and a table that represent the alert count, alert type, and alert percentage appear in the **Alerts for Period** section.
4. Select an alert type that appears in the donut chart or table from the **Alerts for Period** section.

   For the alert type that you select in the **Alerts for Period** section, a graph that represents the number of alerts for a time period appears in the **Execution Failures** section.
5. To view detailed information of an alert that appears in the **Execution Failures** section, click the **View Details** button in **Details** column for that alert.

   The alert details appear in the **Alert Details** pane.

Export Alerts for PowerCenter Operations

You can export alerts for the current day or for a specified period to a standard comma-separated value (CSV) file or an excel (.xlsx) file. The exported file includes a snapshot of the alerts generated for the current

day or the alert history for a specified period. The report depicts the rule categories and the alert records. You can also choose to export the alert body.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Reports** tab.
2. Click the **Operations** tab, and choose if you want to export the alerts for the current day or the alert history.
   - To export alerts from the current day, click **Today** and then click the **Export** icon.
   - To export the alert history, click **History**, use the predefined views or the slider control to select a timeline to display the alerts, and then click the **Export** icon.
     By default, the timeline is set to 1 month.

   The **Export Alerts** dialog box appears.
3. From the menu, choose the format in which you want to export the alerts.
   - Select **Excel** to export the alerts to an .xlsx format.
     The file includes a summary sheet, along with a donut chart that represents the rule group and the alert count for each group. Each rule category sheet displays the object type, rule name, alert time, and other alert details. You can view the report only in Microsoft Office version 2007 and later versions.
   - Select **Csv** to export the alerts to a .csv file.
     The file contains all the alert records aggregated in one sheet.
4. To export the details of the alert, select **Include Alert Details**.
5. Click **Save**.

Monitoring PowerCenter Governance Alerts

Monitor the PowerCenter development environment alerts from the **Governance** tab on the **Reports** tab. You can view a comprehensive report or drill down into specific alert details of the current and historical alerts for PowerCenter governance.

The contents that appear on the **Governance** tab vary based on the view that you select.

You can select the following views:

- **Today**. View and export the PowerCenter development environment alerts for the current day.
- **History**. View and export the PowerCenter development environment alerts for a particular time period.
- **On Demand Reports**. Create and view on demand reports for workflows, sessions, or transformations in PowerCenter development environment.

Monitoring the PowerCenter Governance Alerts for the Current Day

You can monitor the PowerCenter governance alerts for the current day in the **Today** view on the **Governance** tab.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Reports** tab.
2. Click the **Governance** tab and then click the **Today** view. The following sections appear in the contents panel:

Total Alerts

The alert type, alert count, and percentage of the alert types appear in a graph and table format in the **Total Alerts** section.

Best Practice Violations

The alert details appear in the **Best Practice Violations** section.

3. To view the best practice violations of a particular alert type, click on the corresponding alert type from the donut chart or table. The violations for the alert type that you select appear in the **Best Practice Violations** section.
4. To view detailed information of an alert that appears in the **Best Practice Violations** section, click the **View Details** button in **Details** column for that alert.

Monitoring the PowerCenter Governance Best Practice Violations

You can monitor the PowerCenter governance best practice violations for a particular time period in the **History** view on the **Governance** tab.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Reports** tab.
2. Click the **Governance** tab and then click the **History** view.

   The **Total Alerts**, **Alerts for Period**, and **Best Practice Violations** sections appear in the contents panel. A graph that represents the number of alerts over a period of time appears in the **Total Alerts** section.
3. Select the time period or alert type for which you want display the best practice violations.
   - To monitor best practice violations for a time period, select the time period with the slider or views in the **Total Alerts** section.
   - To monitor best practice violations based on the alert type, select an alert type that appears in the donut chart or table from the **Alerts for Period** section.

   A graph that represents the number of best practice violations and the list of best practice violations for the time period appears in the **Best Practice Violations** section.
4. To view detailed information of an alert that appears in the **Best Practice Violations** section, click the **View Details** button in **Details** column for that alert.

   The alert details appear in the **Alert Details** pane.

Export Alerts for PowerCenter Governance

You can export alerts for the current day or for a specified period to a standard comma-separated value (CSV) file or an excel (.xlsx) file. The exported file includes a snapshot of the alerts generated for the current day or the alert history for a specified period. The report depicts the rule categories and the alert records. You can also choose to export the alert body.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Reports** tab.
2. Click the **Governance** tab, and choose if you want to export the alerts for the current day or the alert history.
   - To export alerts from the current day, click **Today** and then click the **Export** icon.
   - To export the alert history, click **History**, use the predefined views or the slider control to select a timeline to display the alerts, and then click the **Export** icon.
     By default, the timeline is set to 1 month.

   The **Export Alerts** dialog box appears.

3. From the menu, choose the format in which you want to export the alerts.
   - Select **Excel** to export the alerts to an .xlsx format.
     The file includes a summary sheet, along with a donut chart that represents the rule group and the alert count for each group. Each rule category sheet displays the object type, rule name, alert time, and other alert details. You can view the report only in Microsoft Office version 2007 and later versions.
   - Select **Csv** to export the alerts to a .csv file.
     The file contains all the alert records aggregated in one sheet.
4. To export the details of the alert, select **Include Alert Details**.
5. Click **Save**.

Monitoring the PowerCenter Governance Alerts Using On Demand Reports

Use the on demand reports to retrieve information on instances of sessions, transformations, or workflows for an attribute value. You can create and view on demand reports in the **On Demand Reports** view on the **Governance** tab.

Use the **On Demand Reports** view to create and save on demand reports, edit the saved reports, and run on demand reports. For example, you can create an on demand report to retrieve all sessions for an attribute with a specified attribute value.

Creating On Demand Reports

You can create on demand reports from the **On Demand Reports** view on the **Governance** tab.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Reports** tab.
2. Click the **Governance** tab and then click the **On Demand Reports** view.
3. In the **Actions** menu on the **Reports** tab, click **New**.

   The **Profiles** screen appears.
4. Enter a name for the on demand report profile.
5. Optionally, enter a description for the on demand profile.
6. Select the Repository Service that manages the PowerCenter development environment.

   The folders associated with that Repository Service appear in the **Folder** list.
7. Select the folder in which the session, workflow, or transformation objects exist.
8. Select the workflow, session, or transformation object type that you want to monitor.

   The attributes specific to the selected object types appears in the **Attribute Check** list.
9. Select the attribute that you want to check from the **Attribute Check** list, and then specify the condition and value that you want to check.

   For example, if the attribute name is Commit Type, and the decoded attribute value is Source, then use the attribute value as 0.
10. Click **Save**.

Editing On Demand Reports

You can edit on demand reports that you create from the **On Demand Reports** view on the **Governance** tab.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Reports** tab.
2. Click the **Governance** tab and then click the **On Demand Reports** view.
3. Select the report profile that you want to edit from the **Profiles** section.
4. In the **Actions** menu on the **Reports** tab, click **Edit**.

   The **Profiles** screen appears.
5. Edit the on demand report properties.

   The following table displays the on demand report properties:

Property	Description
Name	Name of the on demand report profile.
Description	Description of the on demand report profile.
Repository Service	Repository Service that manages the PowerCenter development environment.
Folder	Folder in which the session, workflow, or transformation objects exist.
Artifact Type	Type of the object that you want to monitor.
Attribute Check	Attribute that you want to check for the selected object.

6. Click **Save**.

Deleting On Demand Reports

You can delete on demand reports from the **On Demand Reports** view on the **Governance** tab.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Reports** tab.
2. Click the **Governance** tab and then click the **On Demand Reports** view.
3. Select the report profile that you want to delete from the **Profiles** section.
4. In the **Actions** menu on the **Reports** tab, click **Delete**.

Running On Demand Reports

You can run on demand reports from the **On Demand Reports** view on the **Governance** tab.

1. In the Proactive Monitoring for PowerCenter Management Console, click the **Reports** tab.
2. Click the **Governance** tab and then click the **On Demand Reports** view.
3. Select the report profile that you want to run from the **Profiles** section.
4. In the **Actions** menu on the **Reports** tab, click **Run**.

   The **Run Profile** screen appears.
5. Enter the user name of the PowerCenter user.
6. Select the start date for the period for which you want to run the report. The format for the date is in yyyy-mm-dd hh:mm:ss format.

7. Click the **Run** button.

   The **Report** screen appears with a report for the selected workflow, session, or transformation object.

Attribute Values for On-Demand Reports

The following table provides attribute values and descriptions for each attribute name that you can configure:

Attribute Name	Attribute Value and Description
Commit Type	Use one of the following values: - 0. Source - 1. Target - 2. User-defined
Decimal Separator	Use one of the following values: - 1. comma (,) - 2. period (.)
Dynamic Partitioning	Use one of the following values: - 0. Disabled - 1. Based on the number of partitions - 2. Based on the number of nodes in a grid - 3. Based on source partitioning - 4. Based on the number of CPUs
Error Log Type	Use one of the following values: - 0. None - 1. Relational database - 2. Flat file
Is Partitionable	Use one of the following values: - 0. No - 1. Locally - 2. Across a grid
Join Type	Use one of the following values: - 1. Normal join - 2. Master outer join - 3. Detail outer join - 4. Full outer join
Lookup Policy on Multiple Match	Use one of the following values: - 1. Use first value - 2. Use last value - 3. Report error - 4. Use any value
Master Sort Order	Use one of the following values: - 0. Ascending - 1. Auto
Null Ordering	Use one of the following values: - 0. Null is highest value - 1. Null is lowest value
Null Ordering in detail	Use one of the following values: - 0. Null is highest value - 1. Null is lowest value

Attribute Name	Attribute Value and Description
Null Ordering in master	Use one of the following values: - 0. Null is highest value - 1. Null is lowest value
On Pre-Post SQL error	Use one of the following values: - 0. Stop - 1. Continue
On Pre-session command task error	Use one of the following values: - 0. Stop - 1. Continue
On Stored Procedure error	Use one of the following values: - 0. Stop - 1. Continue
Output Is Repeatable	Use one of the following values: - 0. Never - 1. Based on input order - 2. Always
Override Tracing	Use one of the following values: - 0. None - 1. Terse - 2. Normal - 3. Verbose initialization - 4. Verbose data
Pre-build Lookup Cache	Use one of the following values: - 0. Auto - 1. Always allow - 2. Never allow
Pushdown Optimization	Use one of the following values: - 0. None - 1. To source - 2. To target - 3. Full - 4. $PushdownConfig
Recovery Strategy	Use one of the following values: - 0. Fail task and continue workflow - 1. Resume from last checkpoint - 2. Restart task
Save Session log by	Use one of the following values: - 0. By runs - 1. By timestamp
Save Workflow log by	Use one of the following values: - 0. By runs - 1. By timestamp
Source Type	Use one of the following values: - 1. Database - 2. Flat file - 3. Source qualifier

Attribute Name	Attribute Value and Description
Stored Procedure Type	Use one of the following values: - 1. Target pre-load - 2. Target post-load - 3. Normal - 4. Source pre-load - 5. Source post-load
Thousand Separator	Use one of the following values: - 0. None - 1. Comma (,) - 2. Period (.)
Top/Bottom	Use one of the following values: - 0. Bottom - 1. Top
Tracing Level	Use one of the following values: - 1. Terse - 2. Normal - 3. Verbose initialization - 4. Verbose data
Transformation Scope	Use one of the following values: - 0. Row - 1. Transaction - 2. All input
Treat Source Rows As	Use one of the following values: - 0. Insert - 1. Delete - 2. Update - 3. Data driven
Type	Use one of the following values: - 0. Informatica - 1. COM

CHAPTER 5

Proactive Monitoring SNMP Alerts

This chapter includes the following topic:

- Proactive Monitoring SNMP Alerts, 46

Proactive Monitoring SNMP Alerts

You can configure the PMPC solution to send alerts as SNMP traps in addition to email and RTAM alerts. PMPC solution supports SNMP v2.

Configure the following PMPC components to send alerts as SNMP traps:

- SNMP Rule
- SNMP Responder
- SNMP Response

SNMP Rule

The SNMP rule generates SNMP traps for alerts. The rule activates whenever an event occurs on pc_notifications. The rule takes following variable values from the incoming activation and binds them to the trap Protocol Data Unit (PDU): body, priority, extended_properties, rs, rulename, artifact_type, domain, persona, product, rule_user, rule_group, and subject.

The PC_S7 SNMP Notification Response rule is available as part of RulePoint installation and is disabled by default. To reactivate the rule, edit the PC_S7 SNMP Notification Response rule from RulePoint and change the status of the rule to active.

To add bindings, check for the corresponding variable definition in the MIB. If the variable definition is present in the MIB then add the parameter name to OID mapping to the source code. Add the property to the response parameters.

SNMP Responder

The SNMP responder sends the trap PDU to the network manager that you configure to receive traps. The SNMPv2Responder is available as part of RulePoint installation and is disabled by default. To configure the SNMPv2Responder, edit the SNMPv2Responder from RulePoint and provide the network manger details.

The following table describes the responder properties that you need to configure:

Property	Description
Name	Name for the responder.
NMS Host	IP address of the host on which NMS is available.
NMS Port	Port number on which NMS will receive traps. The default value is 162.
Community String	Community string for SNMPv2. The default values is public.
Timeout	Timeout value for trap in milliseconds. The default values is 0.
Retry Count	Retry count for trap. The default value is 0.
Trap OID	OID of the trap.
Status	Status of the responder. Change the status of the responder to active to reactive the responder.

SNMP Response

The SNMPv2Response is available as part of RulePoint installation and is disabled by default. The SNMPv2Response is the default response for SNMPv2Responder.

Use the response parameters map values to construct the response variables. Edit the SNMPv2Response in Rulepoint. Change the status of the response to active to reactivate the response.

CHAPTER 6

Proactive Monitoring Watchlists

This chapter includes the following topic:

- Proactive Monitoring Watchlists, 48

Proactive Monitoring Watchlists

The following table lists the predefined watchlists that are available by default upon installing Proactive Monitoring for PowerCenter Operations:

Watchlists Name	Description	State
PowerCenter Monitored Folders	The list of PowerCenter folders that are monitored. To receive notifications add your folder names to this watchlist. If the folder is same for multiple PowerCenter repositories, you receive this notification for all configured repositories.	Deployed
PowerCenter Verbose Workflows	The list of PowerCenter workflows for which run-time reports are sent. This reports are sent to the pcadmin persona. If the workflow is same for multiple folders and repository services, you receive the run-time reports for all folders and repository services.	Deployed
PowerCenter Repository Service Database Tablespace Name	The list of tablespace names from the PowerCenter repository database that are monitored. If the tablespace name is same for multiple databases, you can get alerts for all the databases.	Deployed

CHAPTER 7

Proactive Monitoring Topics

This chapter includes the following topic:

- Proactive Monitoring Topics, 49

Proactive Monitoring Topics

The following table lists the predefined topics that are available by default upon installing Proactive Monitoring for PowerCenter Operations:

Topic Name	Description	State
pc_notifications	This topic contains event properties associated with the proactive monitoring notification framework.	Deployed
pc_alert_history_purge_request	This topic contains event properties associated with the alert history purge request. Source: PowerCenter Daily Alert History Purge Requestor	Deployed
pc_pmpc_global_settings	This topic contains event properties associated with global settings framework. Source: PowerCenter Load PMPC Global Settings from Database	Deployed
pc_sessions	This topic contains event properties associated with PowerCenter sessions. Source: PowerCenter Sessions Modified Incremental	Deployed
pc_completed_sessions	This topic contains event properties associated with PowerCenter completed sessions. Source: PowerCenter Completed Sessions	Deployed

Topic Name	Description	State
pc_session_to_workflow_ratio	This topic contains event properties associated with PowerCenter session to workflow ratio. Source: PowerCenter Session to Workflow Ratio	Deployed
pc_runtime_server_stats	This topic contains event properties associated with PowerCenter runtime server statistics. Source: PC_O13 Send run time server statistics	Deployed
pc_missed_workflows	This topic contains event properties associated with missed schedules of PowerCenter workflows. Source: PowerCenter Scheduler Missed Workflows	Deployed
pc_concurrent_workflows	This topic contains event properties associated with the PowerCenter workflow instances running concurrently. Source: PowerCenter Concurrent Workflows	Deployed
pc_ping	This topic contains event properties associated with the health status of the Informatica domain. Source: PowerCenter Web Service Hub Ping Service	Deployed
pc_runtime_session_stats	This topic contains event properties associated with the sessions of scheduled and running PowerCenter workflows. Source: PC_O9 Send run time session statistics	Deployed
pc_runtime_stats	This topic contains event properties for the Informatica domain server, running and scheduled workflows, and the associated sessions and targets. Source: PowerCenter Web Service Hub Runtime Statistics Receiver	Deployed

Topic Name	Description	State
pc_runtime_workflow_stats	This topic contains event properties associated with running and scheduled PowerCenter workflows. Source: PC_O8 Send run time workflow statistics	Deployed
pc_scheduled_workflows	This topic contains event properties associated with the schedules of the PowerCenter workflows. Source: PowerCenter Scheduler Scheduled Workflows Incremental	Deployed
pc_runtime_target_stats	This topic contains event properties associated with the PowerCenter targets. Source: PC_O7 Send run time transformation statistics	Deployed
pc_running_workflows	This topic contains event properties associated with the completed workflows in PowerCenter. Source: PowerCenter Running Workflows	Deployed
pc_completed_workflows	This topic contains event properties associated with the completed workflows in PowerCenter. Source: PowerCenter Completed Workflows	Deployed
pc_failed_workflows	This topic contains event properties associated with failed workflows in PowerCenter. Source: PC_OW6 Publish events for failed workflows	Deployed
pc_running_sessions	This topic contains event properties associated with running sessions in PowerCenter. Source: PowerCenter Running Sessions	Deployed
node_process_property	This topic contains event properties associated with the top 'N' CPU and memory consuming processes on the PowerCenter node. Source: Node Process Monitor	Deployed

Topic Name	Description	State
node_database_property	This topic contains event properties associated with the PowerCenter Repository Service database health status and the tablespace usage. Source: PowerCenter Repository Database Tablespaces	Deployed
node_system_property	This topic contains event properties associated with the total CPU and memory consumption on the PowerCenter Integration Service node. Source: Node System Monitor	Deployed
node_topnstats_table_purge	This topic contains event properties associated with purging the node process property table in the Proactive Monitoring database. Source: Node TopNStats Table Purge	Deployed
pc_completed_sessions_trending	This topic contains event properties associated with the Persistence Framework. Source: PC_OS13 Completed sessions cache for persistence framework	Deployed
pc_completed_sessions_table_purge	This topic contains event properties associated with the Persistence Framework. Source: PowerCenter completed sessions shadow table purge	Deployed
pc_completed_workflows_table_purge	This topic contains event properties associated with the housekeeping of the powercenter completed workflows shadow table in the Proactive Monitoring database. Source: PowerCenter completed workflows shadow table purge	Deployed
pc_completed_workflows_trending	This topic contains event properties associated with the Persistence Framework. Source: PC_OW11 Completed workflow cache for persistence framework	Deployed

Topic Name	Description	State
pc_aggregated_ops_analytics	This topic contains aggregated counts of sessions, workflows, mappings, and transformations. Source: PowerCenter Metadata Counts	Deployed
pc_running_sessions_count	This topic contains event properties associated with PowerCenter running sessions. Source: PowerCenter Running Sessions Count	Deployed
pc_running_sessions_workflows_count	This topic contains event properties associated with PowerCenter running session workflow counts. Source: PowerCenter Running Sessions Workflows Count Report	Deployed
node_agent_lifecycle_monitor	This topic contains event properties associated with the node agent health status. Source: Node Agent Monitor	Deployed
node_process_lifecycle_monitor	This topic contains events for monitored process on a given host that are stopped. Source: Node Process Lifecycle Monitor	Deployed
node_filefolder_property	This topic contains event properties associated with the file and folder monitoring statistics on the PowerCenter node. Source: File-Folder Monitoring Stats Publisher	Draft

CHAPTER 8

Proactive Monitoring Services

This chapter includes the following topics:

- Sources, 54
- Analytics, 58
- Responders, 61

Sources

The sources fetch data from PowerCenter repositories and run-time instances which are used for rule evaluation.

The following table lists the predefined sources that are available by default after installing Proactive Monitoring for PowerCenter Operations:

Source Service Name	Description	Properties	State
Node Agent Monitor	Retrieves the health status of the node agent.	- Type: Interval based SQL Service - Topic: node_agent_lifecycle _monitor - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 300 seconds	Deployed
Node Process Lifecycle Monitor	Retrieves the details of process monitored by node agent.	- Type: Interval based SQL Service - Topic: node_process_lifecycle_monitor - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 300 seconds	Deployed
Node Process Monitor	Retrieves the CPU and memory consuming process information from all the hosts where you have started the node agent.	- Type: Interval based SQL Service - Topic: node_process_property - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 600 seconds	Deployed

Source Service Name	Description	Properties	State
Node System Monitor	Retrieves the total CPU and memory information of all the hosts where you have started the node agent.	- Type: Interval based SQL Service - Topic: node_system_property - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 600 seconds	Deployed
Node TopNStats Table Purge	This is a system source service which is used for purging node statistics table.	- Type: SQL - Topic: node_topnstats_table_purge - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 2 hours	Deployed
PowerCenter Completed Sessions	Retrieve the PowerCenter completed sessions information.	- Type: PMPC SQL Source - Topic: pc_completed_sessions - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 600 seconds	Deployed
PowerCenter completed sessions shadow table purge	This is a system source service used by the Persistence Framework to purge the completed sessions in the shadow table.	- Type: SQL - Topic: pc_completed_sessions_table_purge - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: Daily	Deployed
PowerCenter Completed Workflows	Retrieve the PowerCenter completed workflows information.	- Type: PMPC SQL Source - Topic: pc_completed_workflows - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 600 seconds	Deployed
PowerCenter completed workflows shadow table purge	This is a system source service used by the Persistence Framework to purge the completed workflows in the shadow table.	- Type: SQL - Topic: pc_completed_workflows_table_purge - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: Daily	Deployed
PowerCenter Concurrent Workflows	Retrieve the PowerCenter workflows that have more than one instance running concurrently.	- Type: PMPC SQL Source - Topic: pc_concurrent_workflows - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 600 seconds	Deployed

Source Service Name	Description	Properties	State
PowerCenter Daily Alert History Purge Requestor	Retrieve alert history purge frequency from the global settings framework. The frequency is used for purging the alert history.	- Type: SQL - Topic: pc_alert_history_purge_request - Connected to: Rulepoint Repository (pc_rp) - Default interval: Daily	Deployed
PowerCenter Load PMPC Global Settings from Database	Retrieve Proactive Monitoring global settings from the RulePoint database.	- Type: SQL - Topic: pc_pmpc_global_settings - Connected to: - - Default interval: 21600 seconds	Deployed
PowerCenter Metadata Counts	This source extracts the aggregate count of sessions, workflows, mappings, and transformations from PowerCenter repository and stores it in Proactive Monitoring database.	- Type: PMPC Metadata Receiver - Topic: pc_aggregated_ops_analytics - Connected to: - - Default interval: 21600 seconds	Deployed
PowerCenter Repository Database Tablespaces	Retrieve the database state and tablespace details from PowerCenter repository databases.	- Type: PMPC Database Monitor - Topic: node_database_property - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 10 minutes	Deployed
PowerCenter Running Sessions	Retrieve information of the all running sessions of PowerCenter.	- Type: PMPC SQL Source - Topic: pc_running_sessions - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 600 seconds	Deployed
PowerCenter Running Sessions Count	Retrieve the total number of sessions running in the PowerCenter repository.	- Type: PMPC SQL Source - Topic: pc_running_sessions_count - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 600 seconds	Deployed

Source Service Name	Description	Properties	State
PowerCenter Running Workflows	Retrieve information of all running workflows of PowerCenter.	- Type: PMPC SQL Source - Topic: pc_running_workflows - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 600 seconds	Deployed
PowerCenter Running Sessions Workflows Count Report	Retrieve running sessions and workflows count at a particular point in time	- Type: PMPC SQL Source - Topic: pc_running_sessions_workflows_count - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 600 seconds	Deployed
PowerCenter Scheduler Missed Workflows	Retrieve PowerCenter workflows that missed their schedule.	- Type: PMPC SQL Source - Topic: pc_missed_workflows - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 600 seconds	Deployed
PowerCenter Scheduler Scheduled Workflows Incremental	Retrieve the details of the latest schedule and the next schedule of PowerCenter workflows.	- Type: PMPC SQL Source - Topic: pc_scheduled_workflows - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 600 seconds	Deployed
PowerCenter Session to Workflow Ratio	Retrieve the PowerCenter sessions to workflow ratio.	- Type: PMPC SQL Source - Topic: pc_session_to_workflow_ratio - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 600 seconds	Deployed
PowerCenter Sessions Modified Incremental	Retrieve the details of non-reusable PowerCenter sessions inside a worklet, reusable sessions inside a folder, and non-reusable sessions inside a workflow.	- Type: PMPC SQL Source - Topic: pc_sessions - Connected to: PowerCenter Repository (pcrs_readonly) - Default interval: 21600 seconds	Deployed
PowerCenter Web Service Hub Ping Service	Retrieve the health status of a Informatica domain.	- Type: PMPC WSH Ping Service - Topic: pc_ping - Connected to: PowerCenter Web Services Hub - Default interval: 600 seconds	Deployed

Source Service Name	Description	Properties	State
PowerCenter Web Service Hub Runtime Statistics Receiver	Retrieve the runtime statistics for the Informatica domain server, running and scheduled workflows, and associated sessions and targets.	- Type: PMPC WSH Runtime Statistics Receiver - Topic: pc_runtime_stats - Connected to: PowerCenter Web Services Hub - Default interval: 600 seconds	Deployed
File-Folder Monitoring Stats Publisher	Publishes the monitoring statistics of files and folders on the host that the node agent collects. This source is used for monitoring files or folders. To enable file and folder monitoring, deploy this source along with the rule, PC_021 File-Folder matched the monitoring conditions configured for Host.	- Type: SQL - Topic: node_filefolder_property - Connected to: Rulepoint Repository (pc_rp) - Default interval: 600 seconds	Draft

Analytics

RulePoint analytics implement a data processing function, and it can be referenced in rule activations.

The following table lists the predefined analytics that are available by default after installing Proactive Monitoring for PowerCenter Operations:

Name	Connects to...	Analytic Type	Description	State
get_lp_pc_completed_sessions_csv	RulePoint Repository	SQL	Fetch completed sessions data for reporting purposes	Deployed
get_lp_pc_completed_workflows_csv	RulePoint Repository	SQL	Fetch completed workflows data for reporting purposes.	Deployed

Name	Connects to...	Analytic Type	Description	State
get_no_mapping	PowerCenter Repository	SQL	Obtain the total number of PowerCenter mappings.	Deployed
get_no_rows_lp_pc_completed_sessions	RulePoint Repository	SQL	Get the number of rows from the completed sessions table in the Proactive Monitoring database. This is used by the Persistence Framework.	Deployed
get_no_rows_lp_pc_completed_workflows	RulePoint Repository	SQL	Get the number of rows from the completed workflows table in the Proactive Monitoring database. This is used by the Persistence Framework.	Deployed
get_no_session	PowerCenter Repository	SQL	Obtain the total number of PowerCenter sessions.	Deployed
get_no_transformation	PowerCenter Repository	SQL	Obtain the total number of PowerCenter sessions.	Deployed
get_no_workflow	PowerCenter Repository	SQL	Obtain the total number of PowerCenter workflows.	Deployed
get_previous_no_of_completed_sessions	PowerCenter Repository	SQL	Obtain the total number of preceding runs for a specified PowerCenter session that is completed.	Deployed
get_previous_no_of_completed_workflows	PowerCenter Repository	SQL	Obtain the total number of preceding runs for a specified PowerCenter workflow that is completed.	Deployed
get_running_sessions_count_csv	PowerCenter Repository	PMPC SQL Analytic	Fetch the count for running sessions under a repository at a given point in time, in the csv format	Draft
get_service_resource_stats	RulePoint Repository	PMPC Service Statistics Analytic	Fetches the environment resource stats for PowerCenter node.	Deployed
get_tstamp_aggregated_analytic	RulePoint Repository	SQL	Obtains the time stamp of PowerCenter Metadata Counts.	Deployed
pc_get_alert_history_csv	RulePoint Repository	SQL	Obtain alert history for the previous N days for reporting purposes.	Deployed
pc_get_alert_history_purge_count	RulePoint Repository	SQL	Obtain the count of alert history records to be purged.	Deployed
pc_get_email	RulePoint Repository	SQL	Obtain the email address for a specified recipient of an alert.	Deployed

Name	Connects to...	Analytic Type	Description	State
pc_get_global_setting	RulePoint Repository	SQL	Obtain value from the global settings framework for a specified attribute name.	Deployed
pc_get_purge_cutoff_date	RulePoint Repository	SQL	Obtain the text string for the cutoff date. The cutoff date is specified in the alerts.	Deployed
pc_get_rtam	RulePoint Repository	SQL	Obtain the Real-Time Alert Manager target for a specified recipient of an alert.	Deployed
pc_get_session_count_for_workflow	PowerCenter Repository	PMPC SQL Analytic	Retrieve the number of sessions in a PowerCenter workflow.	Deployed
pc_getPrevNSessionRows	PowerCenter Repository	SQL	Fetch the number of rows loaded in the previous N runs for a specified PowerCenter session.	Deployed
pc_getPrevNSessionTimes	PowerCenter Repository	SQL	Obtain the elapsed time for the previous N runs for a specified PowerCenter session.	Deployed
pc_getPrevNThroughputs	PowerCenter Repository	PMPC SQL Analytic	Fetch the throughput of target for the previous N runs for a specified PowerCenter session.	Draft
pc_getPrevNWorkflowTimes	PowerCenter Repository	SQL	Fetch the elapsed time for the previous N runs for a specified PowerCenter workflow.	Deployed
pc_is_workflow_running	PowerCenter Repository	PMPC SQL Analytic	Obtain the current running status of PowerCenter workflow in a folder.	Deployed
pc_recent_alert	Rulepoint Repository	SQL	Check if a recent alert has been sent for a specific rule name, key-value combination, and a snooze interval. The snooze parameter determines the most recent alert.	Deployed
rs_formatter	Rulepoint Repository	SQL	Format the rs value in the notification response.	Deployed
compare_filefolder_stats	RulePoint Repository	PMPC File-Folder Monitoring Analytic	Fetch the file and folder statistics for the PowerCenter node.	Draft

Responders

With a responder, you can define the interface parameters for a particular type of response, that is the action to be taken when a rule activates. From a single responder, you can create multiple specific responses.

The following table lists the predefined responders that are available by default after installing Proactive Monitoring for PowerCenter Operations:

Responder Service Name	Description	Properties	State
PowerCenter Metadata Counts Responder	Records details obtained from PowerCenter Metadata Counts source.	- Type: SQL Responder - Topic: - - Connected to: Rulepoint Repository (pc_rp)	Deployed
PowerCenter Alert History Purge Responder	Responds to events by purging alert history older than the specified number of days.	- Type: SQL Responder - Topic: - - Connected to: Rulepoint Repository (pc_rp)	Deployed
PowerCenter Alert Recorder	The alert recorder stores details of alerts in the RulePoint database.	- Type: SQL Responder - Topic: - - Connected to: Rulepoint Repository (pc_rp)	Deployed
PowerCenter completed sessions report recorder responder	Responds to events by recording the completed sessions data into the Proactive Monitoring database for reporting.	- Type: SQL Responder - Topic: - - Connected to: Rulepoint Repository (pc_rp)	Deployed
PowerCenter completed workflows report recorder responder	Responds to events by recording the completed workflows data to the Proactive Monitoring database for reporting.	- Type: SQL Responder - Topic: - - Connected to: Rulepoint Repository (pc_rp)	Deployed

Responder Service Name	Description	Properties	State
PowerCenter running session workflow count recorder responder	Responds to events by recording the running session workflow count to the Proactive Monitoring database for reporting.	- Type: SQL Responder - Topic: - - Connected to: Rulepoint Repository (pc_rp)	Deployed
PowerCenter Email Responder	Responds to events by sending email alerts.	- Type: Email - Topic: - - Connected to: Email server	Deployed
PowerCenter Runtime Stats Transformer	Responds to events by transforming them as runtime server, runtime workflow, runtime session, and runtime target events.	- Type: Event Transformer - Topic: pc_runtime_server_stats, pc_runtime_workflow_stats, pc_runtime_session_stats, pc_runtime_target_stats - Connected to: -	Deployed
PowerCenter Notification Responder	Responds to events by transforming them as notification events.	- Type: Event Transformer - Topic: pc_notifications - Connected to: -	Deployed
PowerCenter Web Service Hub Workflow Control Responder	Responds to events by sending start, stop, restart, or abort operation to control the PowerCenter workflow.	- Type: PMPC WSH Workflow Controller - Topic: - - Connected to: PowerCenter Web Services Hub	Deployed
PowerCenter completed workflows recorder responder	Responds to events by recording the completed workflows data to the Proactive Monitoring database.	- Type: SQL - Topic: - - Connected to: -	Deployed
RTAM	Sends notifications to the Real-Time Alert Manager.	- Type: RTAM Responder - Topic: - - Connected to: -	Deployed

Responder Service Name	Description	Properties	State
Process Monitor Table Update Responder	Deletes the records from the na_process_monitor table after generating the alerts.	- Type: SQL Responder - Topic: - - Connected to: -	Deployed
PowerCenter completed sessions and workflows responder	This Responder service will transform events generated by PowerCenter Completed Sessions and Completed Workflows.	- Type: Event Transformer - Topic: - - Connected to: -	Deployed
PowerCenter completed sessions recorder responder	Responds to events by recording the completed sessions data to the Proactive Monitoring database.	- Type: SQL - Topic: - - Connected to: -	Deployed
SNMPv2Responder	Sends SNMP traps as an alert to SNMP Trap Receiver. To enable sending notifications as SNMP traps, deploy this responder along with the rule, PC_S7 SNMP Notification.	- Type: SNMP Responder - Topic: - - Connected to: -	Draft

CHAPTER 9

Proactive Monitoring Templates and Rules

This chapter includes the following topics:

- Proactive Monitoring Templates, 64
- Template Rules, 77
- Advanced Rules, 85

Proactive Monitoring Templates

Templates provide an easier way to create rules. You can create rules from predefined templates. You can restrict input from users by adding simple validations to the template. You can add specific user assistance wherever required to make the use of the template easier.

PC_OST5 Session elapsed time exceeds the recent average

Sends a notification if the session elapsed time exceeds the recent average. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_completed_sessions_trending
Analytics	get_previous_no_of_completed_sessions, getPrevNSessionTimes, get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

The following table describes the template parameters:

Parameter	Description
Minimum Elapsed Minutes	Minimum elapsed minutes for the session.
count	Count of the previous N runs to be used for computing average.
percent	Percentage increase over average.

PC_OWT4 Workflow running time exceeds the repository SLA

Sends a notification if the workflow running time exceeds the specified repository SLA. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_running_workflows
Sources	PowerCenter Running Workflows
Analytics	get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

The following table describes the template parameters:

Parameter	Description
Minimum Elapsed Minutes	Minimum elapsed minutes for the workflow.
num_runs	Number of runs to compute average.
percent	Percentage increase over average.

PC_OST1 Session running time exceeds folder SLA

Notify if the session running time exceeds the folder SLA. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_running_sessions
Sources	PowerCenter Running Sessions
Analytics	get_service_resource_stats

Property	Value
Response	PowerCenter Notification Response
Persona	apparchitect

The following table describes the template parameters:

Parameter	Description
folder	PowerCenter folder name to be monitored for SLA.
sla	Enter the number of minutes that is the SLA for this folder.

PC_OST10 Running Sessions Count exceeds Threshold

Notify if total number of running sessions under a repository exceeds the threshold. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_running_sessions_count
Sources	PowerCenter Running Sessions Count
Analytics	get_running_sessions_count_csv
Response	PowerCenter Notification Response
Persona	pcadmin

The following table describes the template parameters:

Parameter	Description
p_threshold	The threshold value for the number of running sessions.

PC_OST2 Session running time exceeds the repository SLA

Notify if session running time exceeds repository SLA. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_running_sessions
Sources	PowerCenter Running Sessions
Analytics	get_service_resource_stats

Property	Value
Response	PowerCenter Notification Response
Persona	apparchitect

The following table describes the template parameters:

Parameter	Description
sla	Enter the number of minutes that is the SLA for this folder.

PC_OST3 Session running time exceeds specified session SLA

Notify if the session running time exceeds the specified session SLA. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_running_sessions
Sources	PowerCenter Running Sessions
Analytics	get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

The following table describes the template parameters:

Parameter	Description
p_session_name	PowerCenter session name.
p_workflow_name	PowerCenter workflow that contains the session.
p_folder	PowerCenter folder that contains the session.
sla	Enter the number of minutes that is the SLA for this session.

PC_OST4 Session loaded fewer rows than the recent average

Notify if the number of rows loaded for a session are less than the recent average. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_completed_sessions_trending
Sources	PowerCenter Session to Workflow Ratio
Analytics	pc_prev_no_of_completed_sessions, pc_getPrevNSessionRows, pc_get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

The following table describes the template parameters:

Parameter	Description
num_runs	Number of Runs to compute average.
percent	Percent decrease over average.

PC_OST6 Session running time exceeds the recent average

Notify if the session running exceeds the recent average by the specified threshold. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_running_sessions
Sources	PowerCenter Running Sessions
Analytics	prev_no_of_completed_sessions, getPrevNSessionTimes
Response	PowerCenter Notification Response
Persona	apparchitect

The following table describes the template parameters:

Parameter	Description
Minimum Elapsed Minutes	Minimum elapsed minutes for the workflow
num_runs	Number of runs to compute average.
percent	Percentage increase over average.

PC_OST7 Ratio of session elapsed time to the workflow elapsed time is disproportionate

Notify if session to workflow elapsed time ratio is less than the specified threshold. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_session_to_workflow_ratio
Sources	PowerCenter Session to Workflow Ratio
Analytics	pc_session_count_for_workflow, get_service_resource_stats
Response	PowerCenter Notification Response
Persona	datarchitect

The following table describes the template parameters:

Parameter	Description
ratio	Threshold for session to workflow elapsed time.

PC_OST8 Session non-responsive for specified duration

Notify if the session is non-responsive for a specified duration. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_runtime_session_stats
Sources	PowerCenter Web Service Hub Runtime Statistics Receiver
Response	PowerCenter Notification Response, PowerCenter Web Service Hub Workflow Control Response
Persona	pcadmin

The following table describes the template parameters:

Parameter	Description
low_watermark	Number of rows that were processed during the last time window specified. The session is considered non-responsive only if the number of rows processed is less than this value.
p_duration	Enter the time window in minutes for which the above low watermark applies. If you set it to 60, the alert will fire if no more than the low watermark rows were processed during the past hour.

PC_OST9 Session failed after it was last saved

Notify if session has failed after it was saved. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_completed_sessions, pc_sessions
Sources	PowerCenter Completed Sessions, PowerCenter Sessions Modified Incremental
Response	PowerCenter Notification Response
Persona	apparchitect

The following table describes the template parameters:

Parameter	Description
duration	Enter the duration in minutes to watch for correlation between session saved and workflow failed.

PC_OT1 Memory consumption of integration service node exceeds threshold

Notify if the memory usage on the node exceeds the set threshold. The state is draft.

The following table lists the template properties:

Property	Value
Topic	node_system_property
Sources	Node System Monitor
Response	PowerCenter Notification Response
Persona	pcadmin

The following table describes the template parameters:

Parameter	Description
memory_perc_usage	Threshold value for memory usage in percentage.

PC_OT2 Repository service database space consumption exceeds threshold

Notify when tablespace usage exceeds the set threshold. The state is draft.

The following table lists the template properties:

Property	Value
Topic	node_database_property
Sources	Node Database Monitor

Property	Value
Analytics	get_no_mapping, get_no_session, get_no_transformation, get_no_workflow
Response	PowerCenter Notification Response
Persona	pcadmin

The following table describes the template parameters:

Parameter	Description
database_usage_threshold	Threshold value for database tablespace usage in percentage.

PC_OT3 PowerCenter statistics when repository service database space consumption exceeds threshold

Notify with statistics when tablespace usage exceeds the set threshold. The state is draft.

The following table lists the template properties:

Property	Value
Topic	node_database_property
Sources	Node Database Monitor
Analytics	get_no_mapping, get_no_session, get_no_transformation, get_no_workflow
Response	PowerCenter Notification Response
Persona	pcadmin

The following table describes the template parameters:

Parameter	Description
database_usage_threshold	Threshold value for database tablespace usage in percentage.
no_mapping_threshold	Threshold for total number of mappings.
no_session_threshold	Threshold for total number of sessions.
no_transformations_threshold	Threshold for total number of transformations.
no_workflows_threshold	Threshold for total number of workflows.

PC_OT6 CPU consumption of the integration service node exceeds threshold

Notify if the total CPU consumption on the integration service node exceeds the specified threshold. The state is draft.

The following table lists the template properties:

Property	Value
Topic	node_system_property
Sources	Node System Monitor
Response	PowerCenter Notification Response
Persona	pcadmin

The following table describes the template parameters:

Parameter	Description
cpu_perc_usage	Threshold value for CPU usage in percentage.

PC_OT7 CPU utilization for a process on the integration service node exceeds threshold

Notify if CPU utilization for a process running on the integration service exceeds the threshold value. The state is draft.

The following table lists the template properties:

Property	Value
Topic	node_process_property
Sources	Node Process Monitor
Response	PowerCenter Notification Response
Persona	pcadmin

The following table describes the template parameters:

Parameter	Description
cpu_usage_threshold	Threshold value for CPU usage in percentage.

PC_OT8 Memory consumption for a process on the integration service node exceeds threshold

Notify if memory usage for a process running on the integration service exceeds the threshold value. The state is draft.

The following table lists the template properties:

Property	Value
Topic	node_process_property
Sources	Node Process Monitor
Response	PowerCenter Notification Response
Persona	pcadmin

The following table describes the template parameters:

Parameter	Description
memory_usage_threshold	Threshold value for memory usage in percentage.

PC_OWT1 Workflow missed schedule by N minutes

Notify if the workflow misses its schedule by the specified duration. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_missed_workflows
Sources	PowerCenter Scheduler Missed Workflows
Analytics	get_service_resource_stats
Response	PowerCenter Notification Response
Persona	pcadmin

The following table describes the template parameters:

Parameter	Description
minutes	Threshold delay expressed in minutes.

PC_OWT10 Number of concurrent workflows exceeds the specified threshold

Notify if the number of concurrent workflows exceeds the specified threshold. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_concurrent_workflows
Sources	PowerCenter Concurrent Workflows
Response	PowerCenter Notification Response
Persona	pcadmin

The following table describes the template parameters:

Parameter	Description
count	Threshold for concurrent number of workflow instances.

PC_OWT2 Workflow elapsed time exceeds the recent average

Notify if the workflow elapsed time exceeds previous average. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_completed_workflows_trending
Sources	PC_OW11 Completed workflow cache for persistence framework
Analytics	prev_no_of_completed_workflows, PrevNWorkflowTimes, get_service_resource_stats
Response	PowerCenter Notification
Persona	apparchitect

The following table describes the template parameters:

Parameter	Description
Minimum Elapsed Minutes	Minimum elapsed minutes for the workflow.
num_runs	Number of runs to compute average.
percent	Percentage increase over average

PC_OWT3 Running workflow missed schedule

Notify if the running workflow misses its next scheduled run by the specified SLA. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_missed_workflows
Sources	PowerCenter Scheduler Missed Workflows
Analytics	pc_is_workflow_running, get_service_resource_stats
Response	PowerCenter Notification Response
Persona	pcadmin

The following table describes the template parameters:

Parameter	Description
minutes	Threshold delay expressed in minutes.

PC_OWT5 Workflow running time exceeds the recent average

Notify if the workflow running time exceeds the recent average for the specified number of runs. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_running_workflows
Sources	PowerCenter Running Workflows
Analytics	pc_getPrevNWorkflowTimes, get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

The following table describes the template parameters:

Parameter	Description
Minimum Elapsed Minutes	Minimum elapsed minutes for the workflow
num_runs	Number of runs to compute average.
percent	Percentage increase over average.

PC_OWT6 Workflow running time exceeds specified workflow SLA

Notify if the workflow running time exceeds the specified workflow SLA. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_running_workflows
Sources	PowerCenter Running Workflows
Analytics	get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

The following table describes the template parameters:

Parameter	Description
p_workflow_name	PowerCenter workflow name.
p_folder	PowerCenter folder name that contains the workflow.
sla	Number of minutes that is the SLA for this workflow.

PC_OWT7 Workflow running time exceeds the folder SLA

Notify if the workflow running time exceeds the specified folder SLA. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_running_workflows
Sources	PowerCenter Running Workflows
Analytics	get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

The following table describes the template parameters:

Parameter	Description
folder	PowerCenter folder.
sla	Number of minutes that is the SLA for this workflow.

PC_OWT9 Number of errors in the same workflow matches the threshold

Notify if number of errors in a workflow for a specified duration matches the threshold. The state is draft.

The following table lists the template properties:

Property	Value
Topic	pc_failed_workflows
Sources	PC_OW6 Send notification for failed workflows
Response	PowerCenter Notification Response
Persona	apparchitect

The following table describes the template parameters:

Parameter	Description
count	Number of errors that will trigger the alert.
duration	Specify the duration threshold you want to check for.

Template Rules

Template rules are created from templates by specifying values for the parameters. You can customize these template rules based on the requirement.

PC_O14 CPU consumption of the integration service node exceeds 90 percent

Notify if the total CPU consumption of the integration service node exceeds 90 percent. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	node_system_property
Sources	Node System Monitor
Response	PowerCenter Notification Response
Persona	pcadmin

PC_O15 Memory consumption of the integration service node exceeds 90 percent

Notify if the memory consumption of the integration service node exceeds 90 percent. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	node_system_property
Sources	Node System Monitor
Analytics	get_no_running_workflow, get_no_running_task
Response	PowerCenter Notification Response
Persona	pcadmin

PC_O2 CPU utilization for a process on the integration service node exceeds 50 percent

Notify if the CPU utilization of the integration service node exceeds 50 percent. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	node_process_property
Sources	Node Process Monitor
Response	PowerCenter Notification Response
Persona	pcadmin

PC_O3 Memory consumption for a process on the integration service node exceeds 50 percent

Notify if the memory consumption of the integration service node exceeds 50 percent. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	node_process_property
Sources	Node Process Monitor
Response	PowerCenter Notification Response
Persona	pcadmin

PC_O5 Repository service database space consumption exceeds 90 percent

Notify if the repository service database space consumption exceeds 90 percent. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	node_database_property
Sources	PowerCenter Repository Database Tablespaces
Analytics	get_no_mapping, get_no_session, get_no_transformation, get_no_workflow
Response	PowerCenter Notification Response
Persona	pcadmin

PC_OS1 Session running time exceeds average of previous 3 runs by 25 percent

Notify if the session running time exceeds the average of the previous three runs by 25 percent. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_running_sessions
Sources	PowerCenter Running Sessions
Analytics	get_previous_no_of_completed_sessions, pc_getPrevNSessionTimes
Response	PowerCenter Notification Response
Persona	apparchitect

PC_OS11 Session is not responding for the specified SLA of 15 minutes

Notify if a running session does not load any rows for 15 minutes. When this rule is activated, the response restarts the workflow where the session is not responding. Deploy this rule if you want to restart the workflow. State is draft.

The following table lists the rule properties:

Property	Value
Topic	pc_runtime_session_stats
Sources	PowerCenter Web Service Hub Runtime Statistics Receiver
Response	PowerCenter Notification Response, PowerCenter Web Service Hub Workflow Control Response
Persona	pcadmin

PC_OS19 Running Sessions count in any repository is beyond 100

Notify if total number of running sessions under a repository exceeds 100. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_running_sessions_count
Sources	PowerCenter Running Sessions Count
Analytics	get_running_sessions_count_csv
Response	PowerCenter Notification Response
Persona	pcadmin

PC_OS2 Session running time exceeds folder SLA of 6 minutes

Notify if the session running time exceeds the folder SLA of 6 minutes. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_running_sessions
Sources	PowerCenter Running Sessions
Analytics	get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

PC_OS3 Session running time exceeds repository SLA of 10 minutes

Notify if the session running time exceeds the SLA of 10 minutes. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_running_sessions
Sources	PowerCenter Running Sessions
Analytics	get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

PC_OS5 Session loaded 25 percent less rows than the recent average

Notify if the number of rows loaded by a session is 25 percent less than the recent average. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_sess_trending
Sources	PC_OS13 Completed session cache for persistence framework
Analytics	get_previous_no_of_completed_sessions, pc_getPrevNSessionRows, get_service_resource_stats
Response	PowerCenter Notification Response
Persona	dataarchitect

PC_OS6 Ratio of session elapsed time to the workflow elapsed time is less than 0.6

Notify if the ratio of elapsed time for a session to workflow is less than 0.6. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_session_to_workflow_ratio
Sources	Session to Workflow Ratio
Analytics	pc_get_session_count_for_workflow
Response	PowerCenter Notification Response
Persona	dataarchitect

PC_OS7 Session elapsed time exceeds the recent average by 25 percent

Notify if the session elapsed time exceeds the recent average by 25 percent for the monitored folders. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_sessions_trending
Sources	PC_OS13 Completed session cache for persistence framework
Analytics	get_previous_no_of_completed_sessions, pc_getPrevNSessionRows, get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

PC_OS8 Session running time for s_m_load_sports_events exceeds repository SLA of 4 minutes

Notify if the session running time for s_m_load_sports_events exceeds the SLA of 4 minutes. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_running_sessions
Sources	PowerCenter Running Sessions
Analytics	get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

PC_OS9 Session failed within 30 minutes since it was last saved

Notify if a session failed within 30 minutes since it was last saved. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_running_sessions
Sources	PowerCenter Running Sessions
Response	PowerCenter Notification Response
Persona	apparchitect

PC_OW1 Running workflow missed schedule by 3 minutes

Notify if a running workflow missed its schedule by three minutes. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_missed_workflows
Sources	PowerCenter Scheduler Missed Workflows
Analytics	is_workflow_running, get_service_resource_stats
Response	PowerCenter Notification Response
Persona	pcadmin

PC_OW10 Workflow Load Mobile records running more than 2 minutes

Notify if the Workflow Load Mobile records is running more than two minutes. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_running_workflows
Sources	PowerCenter Running Workflows
Analytics	get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

PC_OW13 Three errors in the same workflow within 60 minutes

Notify if three errors occur in the same workflow within 60 minutes. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_failed_workflows
Sources	PC_OW6 Send notification for failed workflows
Response	PowerCenter Notification Response
Persona	apparchitect

PC_OW2 Workflow running time exceeds folder SLA of 5 minutes

Notify if a workflow running time exceeds folder SLA of five minutes. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_running_workflows
Sources	PowerCenter Running Workflows
Analytics	get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

PC_OW3 Workflow running time exceeds repository SLA of 10 minutes

Notify if a workflow running time exceeds repository SLA by 10 minutes. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_running_workflows
Sources	PowerCenter Running Workflows
Analytics	get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

PC_OW4 Workflow running time exceeds recent average by 25 percent

Notify if the workflow running time exceeds the recent average. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_running_workflows
Sources	PowerCenter Running Workflows
Analytics	pc_getPrevNWorkflowTimes, get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

PC_OW5 Scheduled workflow missed the start time by 3 minutes

Notify if the workflow missed its schedule by three minutes. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_missed_workflows
Sources	PowerCenter Scheduler Missed Workflows
Analytics	get_service_resource_stats
Response	PowerCenter Notification Response
Persona	pcadmin

PC_OW8 Number of concurrent workflows exceeds the threshold of two

Notify if the number of concurrent workflows exceeds the threshold of two. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_concurrent_workflows
Sources	PowerCenter Concurrent Workflows
Response	PowerCenter Notification Response
Persona	pcadmin

PC_OW9 Workflow elapsed time exceeds the recent average by 25 percent

Notify if the workflow elapsed time exceeds the recent average by 25 percent. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_workflows_trending
Sources	PC_OW11 Completed workflow cache for persistence framework
Analytics	prev_no_of_completed_workflows, pc_getPrevNWorkflowTimes, get_service_resource_stats
Response	PowerCenter Notification Response
Persona	apparchitect

Advanced Rules

Advanced rules do not have parameters. You can extend these rules once you are comfortable with the functioning of these rules.

PC_GS19 Check for hardcoded file paths in a session

Notify if file paths for sources and targets are hardcoded in the session. You need to deploy this rule to receive the notifications. State is draft.

The following table lists the rule properties:

Property	Value
Topic	pc_sessions
Sources	PowerCenter Sessions Modified Incremental
Analytics	get_session_hardcoded_paths

Property	Value
Response	PowerCenter Notification Response
Persona	apparchitect

PC_O1 Aggregated Ops Analytic Rule

Triggers the PowerCenter Metadata Counts Response to insert the details into the Proactive Monitoring database. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_aggregated_ops_analytics
Sources	PowerCenter Metadata Counts
Response	PowerCenter Metadata Counts Response

PC_O10 Integration service health check

Notify if the integration service is down. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_ping
Sources	PowerCenter Web Service Hub Ping Service
Response	PowerCenter Notification Response
Persona	pcadmin

PC_O11 Repository service health check

Notify if the repository service is down. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_ping
Sources	PowerCenter Web Service Hub Ping Service
Response	PowerCenter Notification Response
Persona	pcadmin

PC_O12 Domain service health check

Notify if the domain service is down. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_ping
Sources	PowerCenter Web Service Hub Ping Service
Response	PowerCenter Notification Response
Persona	pcadmin

PC_O13 Send run time server statistics

Publish the run time server statistics. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_runtime_stats
Sources	PowerCenter Web Service Hub Runtime Statistics Receiver
Response	PowerCenter Runtime Stats Transformer Response

PC_O16 Node Agent health check

Notify if the node agent is down. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	node_agent_lifecycle_monitor
Sources	Node Agent Monitor
Response	PowerCenter Notification Response
Persona	pcadmin

PC_O17 Process Lifecycle Monitor Alert

Notifies if any of the process monitored by the node agent has stopped. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	node_process_lifecycle_monitor
Sources	Node Process Lifecycle Monitor
Response	PowerCenter Notification Response, Process Monitor table update response
Persona	pcadmin

PC_O19 All Integration services health check

Notify if all the integration services under the domain are down. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_ping
Sources	PowerCenter Web Service Hub Ping Service
Response	PowerCenter Notification Response
Persona	pcadmin

PC_O20 All Repository services health check

Notify if all the repository services under the domain are down. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_ping
Sources	PowerCenter Web Service Hub Ping Service
Response	PowerCenter Notification Response
Persona	pcadmin

PC_O21 File-Folder matched the monitoring conditions configured for Host

Notify if the file and folder statistics is matching the host monitoring configuration. This rule is used for monitoring files or folders. To enable file and folder monitoring, deploy this rule along with the source, File-Folder Monitoring Stats Publisher. State is draft.

The following table lists the rule properties:

Property	Value
Topic	node_filefolder_property
Sources	File-Folder Monitoring Stats Publisher
Analytics	compare_filefolder_stats
Response	* Notification Response
Persona	pcadmin

PC_O4 Repository service database health check

Notify if the repository service database is down. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	node_database_property
Sources	Node Database Monitor
Response	PowerCenter Notification Response
Persona	pcadmin

PC_O7 Send run time transformation statistics

Publish the run time transformation statistics. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_runtime_stats
Sources	PowerCenter Web Service Hub Runtime Statistics Receiver
Response	PowerCenter Runtime Stats Transformer Response

PC_O8 Send run time workflow statistics

Publish the run time workflow statistics. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_runtime_stats
Sources	PowerCenter Web Service Hub Runtime Statistics Receiver
Response	PowerCenter Runtime Stats Transformer Response

PC_O9 Send run time session statistics

Publish the run time session statistics. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_runtime_stats
Sources	PowerCenter Web Service Hub Runtime Statistics Receiver
Response	PowerCenter Runtime Stats Transformer Response

PC_OS10 Session contains failed rows

Notify if the completed session contains failed source or target rows. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_sessions
Sources	PowerCenter Completed Sessions
Response	PowerCenter Notification Response
Persona	dataarchitect

PC_OS12 Check if session runs were successful but zero rows were loaded

Notify if the session run status is successful but zero rows are loaded. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_sessions
Sources	PowerCenter Completed Sessions

Property	Value
Response	PowerCenter Notification Response
Persona	dataarchitect

PC_OS13 Completed session cache for persistence framework

Caches the completed session records to the RulePoint repository. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_sessions
Sources	PowerCenter Completed Sessions
Response	PowerCenter Notification Response

PC_OS14 Daily report of completed sessions

Send the completed sessions report for the day. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_sessions_table_purge
Sources	PowerCenter completed sessions shadow table pu
Analytics	get_no_rows_lp_pc_completed_sessions, get_lp_pc_completed_sessions_csv, pc_get_rtam
Response	PowerCenter RTAM Alert
Persona	pcmonitor

PC_OS15 Check whether the Session Source Rejected Records is greater than Zero

Notify if a session has source rejected records greater than zero. You need to deploy this rule to receive the notifications. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_sessions
Sources	PowerCenter Completed Sessions

Property	Value
Response	PowerCenter Notification Response
Persona	dataarchitect

PC_OS15 Check whether the Session Source Rejected Records is greater than Zero

Notify if a session has source rejected records greater than zero. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_sessions
Sources	PowerCenter Completed Sessions
Response	* Notification Response
Persona	dataarchitect

PC_OS16 Check whether the Session Target Rejected Records is greater than Zero

Notify if a session has target rejected records greater than zero. You need to deploy this rule to receive the notifications.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_sessions
Sources	PowerCenter Completed Sessions
Response	PowerCenter Notification Response
Persona	dataarchitect

PC_OS16 Check whether the Session Target Rejected Records is greater than Zero

Notify if a session has target rejected records greater than zero. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_sessions
Sources	PowerCenter Completed Sessions

Property	Value
Response	* Notification Response
Persona	dataarchitect

PC_OS17 Check whether the Session Source Successful Records is equal to Zero

Notifies if a session source successful records is equal to zero. You need to deploy this rule to receive the notifications. State is draft.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_sessions
Sources	PowerCenter Completed Sessions
Response	* Notification Response
Persona	dataarchitect

PC_OS18 Check whether the Session Target Successful Records is equal to Zero

Notify if a session target successful records are less than or equal to zero. You need to deploy this rule to receive the notifications. State is draft.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_sessions
Sources	PowerCenter Completed Sessions
Response	* Notification Response
Persona	dataarchitect

PC_OS20 Completed session cache for reporting framework

Caches the completed session records to the PMPC repository for reporting. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_sessions
Sources	PowerCenter Completed Sessions
Response	PowerCenter completed sessions report recorder response

PC_OS4 Send notification for failed sessions

Notify if the session fails. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_sessions
Sources	PowerCenter Completed Sessions
Response	PowerCenter Notification Response
Persona	pcadmin

PC_OSW1 Running Session Workflow Counts cache for reporting framework

Caches running session workflow counts into PMPC repository for reporting. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_running_sessions_workflows_count
Sources	PowerCenter Running Sessions Workflows Count Report
Response	PowerCenter running session workflow count recorder response

PC_OW11 Completed workflow cache for persistence framework

Caches the completed workflow records to the RulePoint repository. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_workflows
Sources	PowerCenter Completed Workflows
Response	PowerCenter completed workflows recorder

PC_OW12 Daily report of completed workflows

Send the completed workflows report for the day. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_workflows_table_purge
Sources	PowerCenter completed workflows shadow table purge

Property	Value
Analytics	pc_no_rows_lp_pc_completed_workflows, get_lp_pc_completed_workflows_csv, pc_get_rtam
Response	PowerCenter RTAM Alert
Persona	pcmonitor

PC_OW14 Send notification for terminated workflows

Notify if a workflow has terminated. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_workflows
Sources	PowerCenter Completed Workflows
Response	PowerCenter Notification Response
Persona	pcadmin

PC_OW15 Completed workflow cache for reporting framework

Caches the completed workflow records to PMPC repository for reporting. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_workflows
Sources	PowerCenter Completed Workflows
Response	PowerCenter completed workflows report recorder response

PC_OW16 Check for workflows becoming unscheduled

Check if any workflow has become unscheduled. You need to deploy this rule to receive the notifications when the workflow becomes unscheduled. State is draft.

The following table lists the rule properties:

Property	Value
Topic	pc_runtime_workflow_stats
Sources	PC_O8 Send run time workflow statistics

Property	Value
Response	PowerCenter Notification Response
Persona	pcmonitor

PC_OW6 Publish events for failed workflows

Publish events of failed workflows to the pc_failed_workflows topic. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_completed_workflows
Sources	PowerCenter Completed Workflows
Response	PowerCenter Runtime Stats Transformer Response

PC_OW7 Send runtime report for monitored workflow

Send runtime report for monitored workflow. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_runtime_session_stats
Sources	PowerCenter Web Service Hub Runtime Statistics Receiver
Response	PowerCenter Notification Response
Persona	pcadmin

PC_S1 RTAM Notification

Generate RTAM notification. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	Internal

PC_S2 Email Notification

Generate email notification. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	Internal

PC_S3 Daily alert history report

Send a daily report of the alert history. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_alert_history_purge_request
Sources	PowerCenter Daily Alert History Purge Requestor
Analytics	pc_get_alert_history_csv, pc_get_purge_cutoff_date
Response	PowerCenter RTAM Alert
Persona	pcmonitor

PC_S4 Zero records purged from alert history

Send a daily report when zero records are purged. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_alert_history_purge_request
Sources	PowerCenter Daily Alert History Purge Requestor
Analytics	pc_get_alert_history_purge_count, pc_get_purge_cutoff_date
Response	PowerCenter RTAM Alert
Persona	pcmonitor

PC_S5 Purge alert history

Notify when alert history is purged. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_alert_history_purge_request
Sources	PowerCenter Daily Alert History Purge Requestor

Property	Value
Analytics	pc_get_alert_history_purge_count, pc_get_purge_cutoff_date
Response	PowerCenter RTAM Alert
Persona	pcmonitor

PC_S6 Load Global Settings from Database

Generate RTAM alert when global settings are loaded from the database. State is deployed.

The following table lists the rule properties:

Property	Value
Topic	pc_pmpc_global_settings
Sources	PowerCenter Load PMPC Global Settings from Database
Analytics	pc_get_rtam
Response	PowerCenter RTAM Alert
Persona	pcmonitor

PC_S7 SNMP Notification

Generate SNMP trap for alerts. To enable sending notifications as SNMP traps, deploy this rule. State is draft.

The following table lists the rule properties:

Property	Value
Topic	pc_notifications
Response	SNMP v2 Response

CHAPTER 10

Proactive Monitoring Responses

This chapter includes the following topic:

- Proactive Monitoring Responses, 99

Proactive Monitoring Responses

The response is where you define how you want responses if the rule's event matches the rule condition. In addition to simple notification response, such as send an email or text message, you can configure a response to function like an action.

You can configure to send responses to a single user or groups of users through email or Real-Time Alert Manager user interface.

The following table lists the predefined responses that are available by default upon installing Proactive Monitoring for PowerCenter Operations:

Name	Response Type	Description	State
PowerCenter RTAM Alert	RTAM	Sends alerts to Real-Time Alert Manager.	Deployed
PowerCenter Notification Response	Event Transformer	Transforms events to notification events.	Deployed
PowerCenter Email Response	Email	Sends email to the specified users with content based on the response properties.	Deployed
PowerCenter Alert Recorder Response	SQL	Records alerts to the RulePoint database.	Deployed
PowerCenter Alert History Purge Response	SQL	Purges alert history older than the specified number of days.	Deployed

Name	Response Type	Description	State
PowerCenter Web Service Hub Workflow Control Response	PMPC WSH Workflow Controller	Sends start, stop, restart, abort, schedule, unschedule, or recover operation to control the PowerCenter workflow. **Note:** Stop workflow does not abort the workflow but stops the workflow. To abort, use the abort workflow command exclusively.	Deployed
PowerCenter Runtime Stats Transformer Response	Event Transformer	Transforms events to run-time server, run-time workflow, run-time session, and run-time target events.	Deployed
PowerCenter completed sessions recorder response	SQL	Records completed sessions data in Proactive Monitoring database.	Deployed
PowerCenter completed sessions and workflows response	Event Transformer	Transforms events to pc_completed_sessions_trending and pc_completed_workflows_trending.	Deployed
PowerCenter completed workflows recorder response	SQL	Records completed workflows data in Proactive Monitoring database.	Deployed
PowerCenter Metadata Counts Response	SQL	Records aggregated count of sessions, workflows, mappings, and transformations into the Proactive Monitoring database.	Deployed
Process Monitor Table Update Response	SQL	Purges records from na_process_monitor table.	Deployed
PowerCenter completed workflows report recorder response	SQL	Records completed sessions data in Proactive Monitoring database for reports.	Deployed
PowerCenter completed sessions report recorder response	SQL	Records completed sessions data in Proactive Monitoring database for reports.	Deployed
PowerCenter running session workflow count recorder response	SQL	Records running session workflow count data in Proactive Monitoring database for reports	Deployed
SNMP v2 Response	SNMP Response	Sends SNMP traps as an alert to SNMP Trap Receiver. To enable sending notifications as SNMP traps, deploy the rule, PC_S7 SNMP Notification.	Draft

APPENDIX A

Troubleshooting

This appendix includes the following topic:

- Troubleshooting Real-Time Alerts , 101

Troubleshooting Real-Time Alerts

Alerts do not appear in the Real-Time Alert Manager dashboard even though all services are running.

Perform the following tasks to troubleshoot alerts:

Verify that RTAM is configured to receive alerts.

1. Log in to Real-Time Alert Manager as the user that is not receiving alerts.
2. Click the **Settings** tab and select **Global Settings**.
3. In the navigator pane, verify that RTAM is available in the default notification.

Verify that the folders are being monitored.

1. Log in to the Proactive Monitoring Management Console.
2. Click **Manage Objects**.
3. In the navigator pane, click **Monitored Objects** and select **Monitored Folders**.
4. Click **Actions** and select **Add/Remove Monitored Folders**.
5. Add the PowerCenter repository folders that you want to monitor, and click **Save**.
6. You must update the run time option to update the monitored folders. Click **Actions**, select **Update run time**, and click **Save**.

Verify that the responder is deployed.

1. Log in to the RulePoint Console.
2. Click **Design** and select **Responders**. Verify that the status of the responders is in deployed state.

Review the log file.

Review the solutions.log file in the following location: `<Proactive Monitoring installation directory>/bin/logs`

APPENDIX B

Topic Properties Reference

This appendix includes the following topic:

- Topic Properties, 102

Topic Properties

Property	Description
mapping_id	Indicates the internal ID generated within PowerCenter for the mapping.
mapping_is_valid	Indicates whether the mapping status is valid.
version_number	Indicates the version number of the modified PowerCenter mapping within the incremental time window.
tstamp	Indicates the time stamp when the PowerCenter mapping was last saved within the incremental time window.
workflow_name	Indicates the workflow name associated with the PowerCenter mapping.
mapping_name	Indicates the name of this PowerCenter mapping.
rs	Indicates the Repository Service name associated with the PowerCenter mapping.
mapping_last_saved	Indicates the time stamp when the PowerCenter mapping was last saved within the incremental time window.
folder	Indicates the folder name associated with the PowerCenter mapping.
domain	Indicates the domain name associated with the PowerCenter mapping.
mapping_description	Provides the description of the PowerCenter mapping.
user_name	Indicates the name of the user who enforced a change to the PowerCenter mapping within the incremental time window.

The tables list the properties for the topics in Proactive Monitoring for PowerCenter Operations.

The following table lists the properties for the pc_running_sessions_workflows_count topic:

Property	Description
is_name	Indicates the Integration Service name associated with the PowerCenter sessions and workflows that are running.
session_cnt	Indicates the number of sessions running within the Informatica domain.
tstamp	Indicates the snapshot time when the count of running sessions and workflows was extracted.
rs	Indicates the Repository Service name associated with the running PowerCenter sessions and worflows.
workflow_cnt	Indicates the number of workflows that run within the Informatica domain.
domain	Indicates the domain name associated with the running PowerCenter sessions and worflows.

The following table lists the properties for the pc_running_sessions_count topic:

Property	Description
curr_tstamp	Indicates the snapshot time when the count of running sessions was extracted.
workflow_name	Indicates the workflow name associated with the running PowerCenter sessions.
no_running_sessions	Indicates the number of sessions running associated with one single workflow.
rs	Indicates the Repository Service name associated with the running PowerCenter sessions.
folder	Indicates the folder name associated with the running PowerCenter sessions.
domain	Indicates the domain name associated with the running PowerCenter sessions.

The following table lists the properties for the pc_completed_sessions topic:

Property	Description
is_name	Indicates the Integration Service name associated with the completed PowerCenter session.
throughput_rows_per_sec	Indicates the throughput rows per second of the completed PowerCenter session.
curr_tstamp	Indicates the current time of event generation.
session_elapsed_secs	Indicates the elapsed time, in seconds, of the completed PowerCenter session.
successful_source_rows	Indicates the number of successful rows extracted from the source by the completed PowerCenter session.
failed_target_rows	Indicates the number of target failed rows of the completed PowerCenter session.
session_run_status	Indicates the running status of the completed PowerCenter session.

Property	Description
workflow_name	Indicates the workflow name associated with the completed PowerCenter session.
tstamp	Indicates the time stamp when the data is extracted.
session_id	Indicates the internal ID generated within PowerCenter for the session.
last_error_code	Indicates the error code of the last error message encountered within the PowerCenter session.
workflow_run_id	Indicates the workflow run ID of the workflow associated with the completed PowerCenter session.
session_run_status_code	Indicates the running status code of the completed PowerCenter session.
last_error	Indicates the last error message encountered within the PowerCenter session.
session_name	Indicates the name of the completed PowerCenter session.
successful_target_rows	Indicates the number of successful rows pushed to the target by the completed PowerCenter session.
session_start_time	Indicates the start time of the completed PowerCenter session.
session_end_time	Indicates the end time of the completed PowerCenter session.
first_error_code	Indicates the error code of the first error message encountered within the PowerCenter session.
rs	Indicates the Repository Service name associated with the completed PowerCenter session.
workflow_id	Indicates the workflow ID of the workflow associated with the completed PowerCenter session.
folder	Indicates the folder name associated with the completed PowerCenter session.
failed_source_rows	Indicates the number of failed source rows of the completed PowerCenter session.
domain	Indicates the domain name associated with the completed PowerCenter session.

The following table lists the properties for the pc_session_to_workflow_ratio topic:

Property	Description
is_name	Indicates the Integration Service name associated with the PowerCenter session and workflow.
session_name	Indicates the name of the completed PowerCenter session.
session_elapsed_secs	Indicates the elapsed time, in seconds, of the completed PowerCenter session.
session_to_workflow_ratio	Indicates the ratio of the elapsed time, in seconds, of the session with the workflow.

Property	Description
workflow_name	Indicates the name of the completed PowerCenter workflow.
tstamp	Indicates the time stamp when the data is extracted.
workflow_elapsed_secs	Indicates the elapsed time, in seconds, of the completed PowerCenter session.
session_start_time	Indicates the start time of the completed PowerCenter session.
session_end_time	Indicates the end time of the completed PowerCenter session.
rs	Indicates the Repository Service name associated with the completed PowerCenter session and workflow.
workflow_id	Indicates the internal ID generated within PowerCenter for the workflow.
folder	Indicates the folder name associated with the completed PowerCenter session and workflow.
domain	Indicates the domain name associated with the completed PowerCenter session and workflow.

The following table lists the properties for the pc_runtime_server_stats topic:

Property	Description
server_status	Indicates the status of the PowerCenter Integration Service.
server_startup_time	Indicates the start time of the PowerCenter Integration Service.
is_name	Indicates the name of the PowerCenter Integration Service.
server_current_time	Indicates the current time of the PowerCenter Integration Service.
workflow_name	Indicates the name of the PowerCenter workflow.
rs	Indicates the Repository Service name associated with the PowerCenter Integration Service.
folder	Indicates the folder name associated with the PowerCenter Repository Service.
message_subject	Indicates the message subject. In this case, it is - server.
num_workflows	Indicates the number of running workflows.
domain	Indicates the domain name associated with the completed PowerCenter Integration Service.

The following table lists the properties for the pc_missed_workflows topic:

Property	Description
seconds_delayed	Indicates the delayed time unit in seconds of the missed workflow.
is_name	Indicates the integration service name associated with the missed workflow.
next_schedule	Indicates the next schedule of the missed workflow.
tstamp	Indicates the current time stamp.
workflow_name	Indicates the name of the missed workflow.
rs	Indicates the Repository Service name associated with the missed workflow.
workflow_id	Indicates the internal ID generated within PowerCenter for the missed workflow.
latest_schedule	Indicates the current schedule of the missed workflow.
folder	Indicates the folder name associated with the missed workflow.
domain	Indicates the domain name associated with the missed workflow.

The following table lists the properties for the pc_concurrent_workflows topic:

Property	Description
is_name	Indicates the Integration Service name associated with the concurrent workflow.
num_instances	Indicates the number of parallel instances of the concurrent workflow.
workflow_name	Indicates the name of the concurrent workflow.
rs	Indicates the Repository Service name associated with the concurrent workflow.
max_workflow_run_id	Indicates the maximum workflow run ID of the instance within the concurrent workflow.
workflow_id	Indicates the internal ID generated within PowerCenter for the concurrent workflow.
folder	Indicates the folder name associated with the concurrent workflow.
domain	Indicates the domain name associated with the concurrent workflow.

The following table lists the properties for the pc_ping topic:

Property	Description
faulterrorcode	Indicates the error code when you ping PowerCenter services.
faultstring	Indicates the error message when you ping PowerCenter services.
is_name	Indicates the name of the PowerCenter Integration Service.

Property	Description
ping_rs	Indicates the status message when you ping the PowerCenter Repository Service.
ping_is	Indicates the status message when you ping the PowerCenter Integration Service.
workflow_name	Indicates the name of the PowerCenter workflow.
tstamp	Indicates the current time stamp.
rs	Indicates the name of the PowerCenter Repository Service.
ping_domain	Indicates the status message when pinging Informatica domain.
folder	Indicates the folder name associated with the PowerCenter Repository Service.
domain	Indicates the name of the Informatica domain.

The following table lists the properties for the pc_runtime_session_stats topic:

Property	Description
workflow_run_error_msg	Indicates the error message of the PowerCenter workflow associated with the PowerCenter session.
is_name	Indicates the integration service name associated with the PowerCenter session.
session_source_success_rows	Indicates the number of rows extracted successfully from the source by the PowerCenter session.
session_transform_errors	Indicates the number of transformation errors within the PowerCenter session.
session_index	Indicates the internal chronological order of the session when extracted by the PowerCenter Web Service RunTime statistics source.
session_mapping	Indicates the mapping name associated with the PowerCenter session.
session_run_error_code	Indicates the error code of the PowerCenter session run.
session_first_error_code	Indicates the error code of the first error within the PowerCenter session run.
workflow_name	Indicates the workflow name associated with the PowerCenter session.
workflow_end_time	Indicates the workflow end time associated with the PowerCenter session.
workflow_run_type	Indicates the workflow run type associated with the PowerCenter session.
session_log_file	Indicates the absolute log file name of the PowerCenter session.
session_workflow	Indicates the workflow name associated to the PowerCenter session.
session_first_error_msg	Indicates the first error message of the PowerCenter session.

Property	Description
workflow_run_error_code	Indicates the error code of the error message of PowerCenter workflow associated with the PowerCenter session.
rs	Indicates the Repository Service name associated with the PowerCenter session.
session_target_success_rows	Indicates the number of rows loaded successfully to the target by the PowerCenter session.
folder	Indicates the folder name associated with the PowerCenter session.
session_run_error_msg	Indicates the error message of the PowerCenter session run.
server_startup_time	Indicates the start time of the PowerCenter Integration Service.
workflow_status	Indicates the status of the workflow associated with the PowerCenter session.
workflow_elapsed_secs	Indicates the total elapsed time, in seconds, of the workflow associated with the PowerCenter session.
session_target_failed_rows	Indicates the number of failed rows not loaded to the target by the PowerCenter session.
workflow_run_id	Indicates the workflow run ID of the workflow associated with the PowerCenter session.
session_source_failed_rows	Indicates the number of failed rows not extracted by the source of the PowerCenter session.
server_status	Indicates the status of the PowerCenter Integration Service.
server_current_time	Indicates the current time stamp of the PowerCenter Integration Service.
workflow_run_instance_name	Indicates the name of the instance of the workflow associated with the PowerCenter session.
workflow_start_time	Indicates the start time of the workflow associated with the PowerCenter session.
session_folder	Indicates the folder name associated with the PowerCenter session.
workflow_id	Indicates the internal ID generated for the workflow by PowerCenter, connected to the PowerCenter session.
session_status	Indicates the run status of the PowerCenter session.
message_subject	Indicates the message subject. In this case, it is - session.
workflow_log_file_path	Indicates the absolute log file name of the workflow associated with the PowerCenter session.
workflow_index	Indicates the internal chronological order of the workflow when extracted by the PowerCenter Web Service run-time statistics source.
domain	Indicates the name of the Informatica domain associated with the PowerCenter session.

Property	Description
session_num_targets	Indicates the number of targets in a PowerCenter session.
session_instance	Indicates the PowerCenter session instance name.

The following table lists the properties for the pc_runtime_stats topic:

Property	Description
is_name	Indicates the Integration Service name.
session_transform_errors	Indicates the number of transformation errors within the PowerCenter session.
target_last_error_targetmessage	Indicates the last error message encountered by the target within the PowerCenter session.
session_index	Indicates the internal chronological order of the session when extracted by the PowerCenter Web Service run-time statistics source.
session_mapping	Indicates the mapping name associated with the PowerCenter session.
workflow_end_time	Indicates the workflow end time associated with the PowerCenter session.
workflow_run_type	Indicates the workflow run type associated with the PowerCenter session.
target_type	Indicates the type of the target in the mapping associated with the PowerCenter session. For example, the target can be Relational or Flat File.
rs	Indicates the Repository Service name.
target_rejected_rows	Indicates the number of rejected rows at the target of the PowerCenter session.
session_target_success_rows	Indicates the number of rows loaded successfully to the target by the PowerCenter session.
folder	Indicates the folder name associated with the PowerCenter session.
session_run_error_msg	Indicates the error message of the PowerCenter session run.
session_workflow	Indicates the workflow name associated to the PowerCenter session.
target_last_error_code	Indicates the last error code in the target of the PowerCenter session.
workflow_status	Indicates the status of the workflow associated with the PowerCenter session.
workflow_elapsed_secs	Indicates the elapsed time, in seconds, of the completed PowerCenter workflow.
session_target_failed_rows	Indicates the number of failed rows not loaded to the target by the PowerCenter session.
num_workflows	Indicates the number of workflows running.

Property	Description
workflow_run_id	Indicates the workflow run ID of the workflow associated with the PowerCenter session.
target_end_time	Indicates the end time of the target execution in the PowerCenter session.
workflow_run_instance_name	Indicates the name of the instance of the workflow associated with the PowerCenter session.
workflow_index	Indicates the internal chronological order of the workflow when extracted by the PowerCenter Web Service run-time statistics source.
workflow_log_file_path	Indicates the absolute log file name of the workflow associated with the PowerCenter session.
session_instance	Indicates the PowerCenter session instance name.
workflow_run_error_msg	Indicates the error message of the PowerCenter workflow associated with the PowerCenter session.
session_source_success_rows	Indicates the number of rows extracted successfully from the source by the PowerCenter session.
session_first_error_code	Indicates the error code of the first error within the PowerCenter session run.
session_run_error_code	Indicates the error code of the PowerCenter session run.
workflow_name	Indicates the name of the PowerCenter workflow.
target_name	Indicates the name of the target in the PowerCenter session.
session_log_file	Indicates the absolute log file name of the PowerCenter session.
target_throughput	Indicates the throughput rows per second of the target in the PowerCenter session.
session_first_error_msg	Indicates the first error message of the PowerCenter session.
workflow_run_error_code	Indicates the error code of the error message of PowerCenter workflow associated with the PowerCenter session.
target_index	Indicates the internal chronological order of the target within the PowerCenter session when extracted by the PowerCenter Web Service RunTime statistics source.
server_startup_time	Indicates the start time of the PowerCenter Integration Service.
target_instance_name	Indicates the name of the target instance in the PowerCenter session.
target_affected_rows	Indicates the number of loaded rows successfully to the target of the PowerCenter session.
target_applied_rows	Indicates the number of rows pushed to the target of the PowerCenter session.

Property	Description
session_source_failed_rows	Indicates the number of failed rows not extracted by the source of the PowerCenter session.
server_status	Indicates the status of the PowerCenter Integration Service.
server_current_time	Indicates the current time of the PowerCenter Integration Service.
workflow_start_time	Indicates the start time of the workflow associated with the PowerCenter session.
session_folder	Indicates the folder name associated with the PowerCenter session.
target_start_time	Indicates the start time of the target execution in the PowerCenter session.
message_subject	Indicates the message subject. For example, the message subject can be server or session.
session_status	Indicates the run status of the PowerCenter session.
domain	Indicates the name of the Informatica domain.
session_num_targets	Indicates the number of targets in a PowerCenter session.
target_elapsed_secs	Indicates the elapsed time, in seconds, of the target within the PowerCenter session.

The following table lists the properties for the pc_runtime_workflow_stats topic:

Property	Description
workflow_run_error_msg	Indicates the error message of the PowerCenter workflow.
server_startup_time	Indicates the start time of the PowerCenter Integration Service.
is_name	Indicates the integration service name associated with the PowerCenter workflow.
workflow_name	Indicates the PowerCenter workflow name.
workflow_status	Indicates the status of the PowerCenter workflow.
workflow_elapsed_secs	Indicates the elapsed time, in seconds, of a completed PowerCenter session.
workflow_end_time	Indicates the end time of the PowerCenter workflow execution.
workflow_run_id	Indicates the internal run ID generated for the PowerCenter workflow.
workflow_run_type	Indicates the run type of the PowerCenter workflow.
server_status	Indicates the status of the PowerCenter Integration Service.
server_current_time	Indicates the current time of the PowerCenter Integration Service.
workflow_run_instance_name	Indicates the name of the instance of the PowerCenter workflow.

Property	Description
workflow_start_time	Indicates the start time of the PowerCenter workflow execution.
workflow_run_error_code	Indicates the error code of the error message of PowerCenter workflow.
rs	Indicates the Repository Service name associated with the PowerCenter workflow.
message_subject	Indicates the message subject. In this case, it is - workflow.
folder	Indicates the folder name associated with the PowerCenter workflow.
workflow_log_file_path	Indicates the absolute log file name of the PowerCenter workflow.
workflow_index	Indicates the internal chronological order of the workflow when extracted by the PowerCenter Web Service RunTime statistics source.
domain	Indicates the name of the Informatica domain associated with the PowerCenter workflow.

The following table lists the properties for the pc_scheduled_workflows topic:

Property	Description
seconds_delayed	Indicates the delayed time unit in seconds of the scheduled PowerCenter workflow.
is_name	Indicates the integration service name associated with the scheduled PowerCenter workflow.
wflow_latest_schedule	Indicates the current schedule of the scheduled PowerCenter workflow.
workflow_name	Indicates the name of the scheduled PowerCenter workflow.
tstamp	Indicates the current time stamp.
rs	Indicates the Repository Service name associated with the scheduled PowerCenter workflow.
workflow_id	Indicates the internal ID generated within PowerCenter for the scheduled workflow.
latest_schedule	Indicates the current schedule of the scheduled PowerCenter workflow.
folder	Indicates the folder name associated with the scheduled PowerCenter workflow.
is_id	Indicates the internal ID generated within PowerCenter for the Integration Service.
domain	Indicates the name of the Informatica domain associated with the scheduled PowerCenter workflow.
wflow_next_schedule	Indicates the next schedule of the scheduled PowerCenter workflow.

The following table lists the properties for the pc_runtime_target_stats topic:

Property	Description
workflow_run_error_msg	Indicates the error message of the PowerCenter workflow associated with the PowerCenter session.
is_name	Indicates the Integration Service name associated with the PowerCenter session.
session_source_success_rows	Indicates the number of rows extracted successfully from the source by the PowerCenter session.
session_transform_errors	Indicates the number of transformation errors within the PowerCenter session.
target_last_error_targetmessage	Indicates the last error message encountered by the target within the PowerCenter session.
session_index	Indicates the internal chronological order of the session when extracted by the PowerCenter Web Service RunTime statistics source.
session_mapping	Indicates the mapping name associated with the PowerCenter session.
session_run_error_code	Indicates the error code of the PowerCenter session run.
session_first_error_code	Indicates the error code of the first error within the PowerCenter session run.
workflow_name	Indicates the name of the PowerCenter workflow associated with the PowerCenter session.
target_name	Indicates the name of the target in the PowerCenter session.
workflow_end_time	Indicates the workflow end time associated with the PowerCenter session.
workflow_run_type	Indicates the workflow run type associated with the PowerCenter session.
target_type	Indicates the type of the target in the mapping associated with the PowerCenter session. For example, the target type can be Relational or Flat File.
target_throughput	Indicates the throughput rows per second of the target in the PowerCenter session.
session_log_file	Indicates the absolute log file name of the PowerCenter session.
session_first_error_msg	Indicates the first error message of the PowerCenter session.
workflow_run_error_code	Indicates the error code of the error message of PowerCenter workflow associated with the PowerCenter session.
rs	Indicates the Repository Service name associated with the PowerCenter session.
target_rejected_rows	Indicates the number of rejected rows at the target of the PowerCenter session.
session_target_success_rows	Indicates the number of rows loaded successfully to the target by the PowerCenter session.
folder	Indicates the folder name associated with the PowerCenter session.

Property	Description
target_index	Indicates the internal chronological order of the target within the PowerCenter session when extracted by the PowerCenter Web Service RunTime statistics source.
session_run_error_msg	Indicates the error message of the PowerCenter session run.
server_startup_time	Indicates the start time of the PowerCenter Integration Service associated with the PowerCenter session.
target_instance_name	Indicates the name of the target instance in the PowerCenter session.
target_last_error_code	Indicates the last error code in the target of the PowerCenter session.
workflow_status	Indicates the status of the workflow associated with the PowerCenter session.
workflow_elapsed_secs	Indicates the elapsed time, in seconds, of the completed PowerCenter workflow associated with the PowerCenter session.
session_target_failed_rows	Indicates the number of failed rows not loaded to the target by the PowerCenter session.
target_affected_rows	Indicates the number of rows loaded successfully to the target of the PowerCenter session.
target_applied_rows	Indicates the number of rows pushed to the target of the PowerCenter session.
workflow_run_id	Indicates the workflow run ID of the workflow associated with the PowerCenter session.
session_source_failed_rows	Indicates the number of failed rows not extracted by the source of the PowerCenter session.
server_status	Indicates the status of the PowerCenter Integration Service associated with the PowerCenter session.
target_end_time	Indicates the end time of the target execution in the PowerCenter session.
server_current_time	Indicates the current time stamp of the PowerCenter Integration Service associated with the PowerCenter session.
workflow_run_instance_name	Indicates the run instance name of the PowerCenter workflow associated with the PowerCenter session.
workflow_start_time	Indicates the start time of the workflow associated with the PowerCenter session.
target_start_time	Indicates the start time of the target execution in the PowerCenter session.
session_status	Indicates the run status of the PowerCenter session.
message_subject	Indicates the message subject. In this case, it is - target.
workflow_log_file_path	Indicates the absolute log file name of the workflow associated with the PowerCenter session.

Property	Description
workflow_index	Indicates the internal chronological order of the workflow when extracted by the PowerCenter Web Service RunTime statistics source.
domain	Indicates the domain name associated with the completed PowerCenter session.
session_num_targets	Indicates the number of targets in a PowerCenter session.
session_instance	Indicates the PowerCenter session instance name.
target_elapsed_secs	Indicates the elapsed time, in seconds, of the target within the PowerCenter session.

The following table lists the properties for the pc_running_workflows topic:

Property	Description
workflow_run_error_msg	Indicates the error message of the running PowerCenter workflow.
is_name	Indicates the Integration Service name associated with the running PowerCenter workflow.
curr_tstamp	Indicates the snapshot time when the information about running workflows were extracted.
workflow_name	Indicates the name of the running PowerCenter workflow.
workflow_elapsed_secs	Indicates the elapsed time, in seconds, of the running PowerCenter workflow.
workflow_run_id	Indicates the internal ID generated for every run of the PowerCenter workflow.
server_current_time	Indicates the current time of the PowerCenter Integration Service.
workflow_start_time	Indicates the start time of the running PowerCenter workflow.
workflow_run_error_code	Indicates the error code of the error message in the running PowerCenter workflow.
rs	Indicates the Repository Service name associated with the running PowerCenter workflow.
workflow_id	Indicates the internal ID generated for the PowerCenter workflow.
folder	Indicates the folder name associated with the running PowerCenter workflow.
domain	Indicates the domain name associated with the running PowerCenter workflow.
workflow_log_file_path	Indicates the absolute log file name of the running PowerCenter workflow.

The following table lists the properties for the pc_completed_workflows topic:

Property	Description
workflow_run_status	Indicates the status of the completed PowerCenter workflow run.
is_name	Indicates the Integration Service name associated with the completed PowerCenter workflow.
curr_tstamp	Indicates the snapshot time when the information about completed workflows was extracted.
workflow_run_status_code	Indicates the status code of the completed PowerCenter workflow run.
tstamp	Indicates the snapshot time when the information about completed workflows were extracted.
workflow_name	Indicates the name of the completed PowerCenter workflow.
workflow_elapsed_secs	Indicates the elapsed time, in seconds, of a completed PowerCenter workflow.
workflow_end_time	Indicates the end time of the completed PowerCenter workflow.
workflow_run_id	Indicates the internal ID generated for every run of the PowerCenter workflow.
workflow_start_time	Indicates the start time of the completed PowerCenter workflow.
rs	Indicates the Repository Service name associated with the completed PowerCenter workflow.
workflow_id	Indicates the internal ID generated for the PowerCenter workflow.
folder	Indicates the folder name associated with the completed PowerCenter workflow.
is_id	Indicates the internal ID generated within PowerCenter for the Integration Service associated with the completed PowerCenter workflow.
domain	Indicates the domain name associated with the completed PowerCenter workflow.

The following table lists the properties for the pc_failed_workflows topic:

Property	Description
workflow_run_status	Indicates the status of the PowerCenter workflow run.
is_name	Indicates the Integration Service name associated with the failed PowerCenter workflow.
curr_tstamp	Indicates the snapshot time when the information about failed workflows were extracted.
workflow_run_status_code	Indicates the status code of the PowerCenter workflow run.
tstamp	Indicates the snapshot time when the information about failed workflows were extracted.

Property	Description
workflow_name	Indicates the name of the failed PowerCenter workflow.
workflow_elapsed_secs	Indicates the elapsed time in seconds of the failed PowerCenter workflow.
workflow_end_time	Indicates the end time of the failed PowerCenter workflow.
workflow_run_id	Indicates the internal ID generated for every run of the PowerCenter workflow.
workflow_start_time	Indicates the start time of the failed PowerCenter workflow.
rs	Indicates the Repository Service name associated with the failed PowerCenter workflow.
workflow_id	Indicates the internal ID generated for the PowerCenter workflow.
folder	Indicates the folder name associated with the failed PowerCenter workflow.
is_id	Indicates the internal ID generated within PowerCenter for the Integration Service associated with the failed PowerCenter workflow.
domain	Indicates the domain name associated with the failed PowerCenter workflow.

The following table lists the properties for the pc_running_sessions topic:

Property	Description
session_run_error_msg	Indicates the error message of the PowerCenter session run.
is_name	Indicates the Integration Service name associated with the running PowerCenter session.
curr_tstamp	Indicates the snapshot time when the information about running sessions were extracted.
session_elapsed_secs	Indicates the elapsed time, in seconds, of the running PowerCenter session.
session_run_error_code	Indicates the error code of the PowerCenter session running.
workflow_name	Indicates the name of the PowerCenter workflow associated with the running PowerCenter session.
session_id	Indicates the internal ID generated for the running PowerCenter session.
workflow_run_id	Indicates the internal ID generated for every run of the PowerCenter workflow associated with the running PowerCenter session.
server_current_time	Indicates the current time of the PowerCenter Integration Service associated with the running PowerCenter session.
session_start_time	Indicates the start time of the running PowerCenter session.
rs	Indicates the Repository Service name associated with the running PowerCenter session.
workflow_id	Indicates the internal ID generated for the PowerCenter workflow associated with the running PowerCenter session.

Property	Description
folder	Indicates the folder name associated with the running PowerCenter session.
domain	Indicates the domain name associated with the running PowerCenter session.
session_instance_id	Indicates the internal ID of the running PowerCenter session instance.
session_instance	Indicates the running PowerCenter session instance name.

The following table lists the properties for the node_process_property topic:

Property	Description
is_name	Indicates the Integration Service name associated with the PowerCenter node.
time_elapsed	Indicates the total process elapsed time running on the PowerCenter node.
p_memory	Indicates the total memory of the process running on the PowerCenter node.
process_name	Indicates the name of the process running on the PowerCenter node.
node_name	Indicates the name of the PowerCenter node.
tstamp	Indicates the time stamp during when the statistics was collected.
pid	Indicates the process ID of the process running on the PowerCenter node.
domain	Indicates the domain name associated with the PowerCenter node.
host_name	Indicates the host name on which the PowerCenter node is installed.
p_cpu	Indicates the total CPU of the process running on the PowerCenter node.

The following table lists the properties for the node_database_property topic:

Property	Description
perc_tablespace_usage	Indicates the total percentage of tablespace used in the database.
ping_database	Indicates the status of the database whether it is alive.
tablespace_name	Indicates the tablespace name present in the database.
tstamp	Indicates the time stamp during when the statistics were collected.
rs	Indicates the Repository Service name associated with the PowerCenter node.
total_space_in_tablespace_in_kb	Indicates the total size of tablespace in kilobytes available in the database.

Property	Description
connectionstring	Indicates the connection string to the database.
domain	Indicates the domain name associated with the PowerCenter node.

The following table lists the properties for the node_system_property topic:

Property	Description
is_name	Indicates the Integration Service name associated with the PowerCenter node.
system_no_proc	Indicates the number of processes running on the PowerCenter node.
system_cpu	Indicates the total system CPU of the PowerCenter node.
tstamp	Indicates the time stamp during when the statistics were collected.
domain	Indicates the domain name associated with the PowerCenter node.
host_name	Indicates the host name on which the PowerCenter node is installed.
system_memory	Indicates the total system memory of the PowerCenter node.

The following table lists the properties for the node_filefolder_property topic:

Property	Description
size_percent	Indicates the file or folder size in percentage.
size_kb	Indicates the file or folder size in kilobytes.
is_folder	Indicates whether the entity is a file or a folder.
no_files	Indicates the number of files contained in a folder.
tstamp	Indicates the time stamp during when the statistics was collected.
if_exists	Indicates whether the file or folder exists.
is_empty	Indicates whether the file or folder is empty.
host_name	Indicates the host name on which the file or folder is located.
path	Indicates the location of file or folder.

The following table lists the properties for the node_agent_lifecycle_monitor topic:

Property	Description
repo_tstamp	Indicates the internal repository time stamp updated by the node agent.
current_upd_tstamp	Indicates the internal current time stamp updated by the node agent.
domain	Indicates the domain name associated to the PowerCenter node where the node agent is installed.
host_name	Indicates the host name where the node agent is installed.

The following table lists the properties for the node_process_lifecycle_monitor topic:

Property	Description
p_args	Indicates the process arguments of the monitored process.
process_name	Indicated the name of the process monitored.
tstamp	Indicates the time stamp when the statistics was collected.
pid	Indicates the process ID of the process monitored.
p_endtime	Indicated the end time of the process monitored.
p_starttime	Indicated the start time of the process monitored.
domain	Indicates the domain name associated to the PowerCenter node where the node agent is installed.
host_name	Indicates the host name where the node agent is installed.

APPENDIX C

Frequently Asked Questions

Does Proactive Monitoring solution support out-of-box reporting for alerts?

Yes. Proactive Monitoring solution supports number of out-of-box dashboards for current as well as historical reporting of alerts

What are the mechanisms available in Proactive Monitoring solution to send alerts?

You can use the Proactive Monitoring solution to send alerts as email, RTAM alerts, or SNMP traps or to send to all these systems.

If you use the Proactive Monitoring Management Console to add or remove PowerCenter monitored folders, will the updates appear in the PowerCenter Monitored Folders watchlist in RulePoint?

Yes. The updates made to PowerCenter Monitored Folders in Proactive Monitoring for PowerCenter Management Console will appear in the PowerCenter Monitored Folders watchlist in RulePoint.

Why I do not see data in the Reports dashboard?

Ensure that you schedule the workflows to populate data in Reports dashboard and clear the browser cache.

Why does an on-demand report does not populate data?

Ensure that you run the profile with correct attributes for the session, workflow, or transformation.

Is there any limit on the time period for alert history?

No. The History view will show all alert data that PMPC reports persists.

Why some columns do not display data even when all the services are running?

Ensure that you schedule the workflows to populate data in Reports dashboard and clear the browser cache.

Can Proactive Monitoring for PowerCenter Operations monitor the PowerCenter Integration Service running on a grid?

Yes. The Proactive Monitoring for PowerCenter Operations can monitor the PowerCenter Integration Service that runs on a grid. The node agents that you install on PowerCenter monitoring nodes collect environmental statistics of all hosts of a grid. The solution adds this information in the alerts. For example, the solution detects a workflow with SLA violation on the PowerCenter Integration Service. The solution triggers an alert which includes the CPU and the memory information from all hosts of the grid.

What is a node agent?

A node agent is a Proactive Monitoring solution component that runs as a command line tool on PowerCenter nodes and collects resource usage information from the host machines. The node agent collects the environmental statistics, such as CPU, memory, and process lifecycle details from the host machines. Then, the node agent publishes the information in the Proactive Monitoring repository. The solution uses the information in rule processing and in alerts.

Can a node agent monitor the lifecycle of any process that runs on a PowerCenter host?

Yes. The node agent can monitor the lifecycle of any process running on a PowerCenter host. In the Proactive Monitoring for PowerCenter Management Console, while adding the host properties for configuring the hosts, you can provide the names of the specific processes that you want to monitor. When any process goes down, the solution sends an alert.

How does the RulePoint communicate with the PowerCenter services or database servers?

The RulePoint communicates with the PowerCenter services or database servers through the Proactive_Monitoring user. The RulePoint connects to the PowerCenter repository through a JDBC connection and to the PowerCenter Integration Service through the Web Services Hub WSDL.

How do you determine the frequency of source services?

The alerting frequency of the sources must not be too high or too low. Each SQL source can have a different schedule based on your requirements. You can set the frequency for the SQL source services in the schedules. You can set the frequency of the PMPC SQL Source services in the source configuration.

Can Proactive Monitoring for PowerCenter monitor multiple Informatica domains?

No. You can monitor one domain with an instance of the Proactive Monitoring solution.

Can Proactive Monitoring for PowerCenter Operations monitor multiple nodes in a grid?

Yes. You can monitor multiple nodes in a grid from the Proactive Monitoring for PowerCenter Operations.

How do I configure the Proactive Monitoring sources to connect to the PowerCenter objects?

You can use the Proactive Monitoring Management Console to configure the solution to monitor an Informatica domain. You can use the Management Console to provide details of the nodes and the services that you want to monitor.

The administrator can use the Management Console to configure one or more PowerCenter services for monitoring. The user interface configuration screens in the Management Console replaced the command line utilities, Global Configuration Tool, and the Alert Recipient Tool from the earlier versions of the solution.

To monitor an Informatica domain, configuration through the Management Console is a prerequisite. The administrator provides configuration details of each host and node to the Management Console. To monitor PowerCenter Services, the administrator provides the configuration details of the services.

What are the privileges to enable PowerCenter environment monitoring?

You can monitor an Informatica domain that contains multiple nodes and many application services running on different physical or virtual machines. The application services are PowerCenter Repository Service, PowerCenter Integration Service, and Web Services Hub. You can configure the nodes for grid and high availability setup within a single domain.

You can collect the environmental statistics, such as CPU, memory, and process lifecycle details from the host machines through the node agents that you install on each PowerCenter monitoring node. Then, the node agents publish the information in the Proactive Monitoring repository.

You must run the node agent as a root user or as an administrator on the node that you want to monitor. Set the JAVA_HOME in each node before you start the node agent.

Can I get an alert if the node agent is down?

If the node agent on a monitored node fails to respond with the statistics details from the node, the Proactive Monitoring triggers the *PC_O16 Node Agent health check* rule. This rule alerts the *pcadmin* persona that the node agent running on the particular node is not available.

Why do I not see any alerts in RTAM even when the rules are activated?

You might not be able to receive alerts if the objects "PowerCenter Notification Responder" and "PowerCenter Notification Response" are not deployed. You need to deploy these objects to receive the alerts.

APPENDIX D

Glossary

Analytic

A service that implements a data processing function. An example of an Analytic is a match function that analyzes a set of input elements and returns a true or false if all elements match specific criteria. RulePoint offers a pre-defined set of Analytics. You can add additional Analytics to the system using the RulePoint SDK.

event

A piece of data that is pulled or pushed into RulePoint from a variety of sources. Events can be anything that you have deemed of interest, such as 911 dispatches, breaking news headlines, banking transactions, or persons of interest entering a predefined location.

event set

A grouping of multiple events into a single entity so that RulePoint can process the events at the same time.

event specific timestamp

This timestamp is used for events that have timestamp values as part of their source data. It does not pertain to event timestamp values that you create in the RulePoint database.

Informatica domain

A collection of nodes and services that define the Informatica platform. You group nodes and services in a domain based on administration ownership.

Integration Service

An application service that runs data integration workflows and loads metadata into the Metadata Manager warehouse.

Integration Service process

A process that accepts requests from the PowerCenter Client and from *pmcmd*. The Integration Service process manages workflow scheduling, locks and reads workflows, and starts DTM processes.

node

A logical representation of a machine or a blade. Each node runs a Service Manager that performs domain operations on that node.

node agent

The node agent is a Proactive Monitoring utility that you copy and run on each monitored PowerCenter node. The node agent connects to the Proactive Monitoring repository database through a JDBC connection. The

node agent then collects statistics from the host and stores the details in the Proactive Monitoring repository database.

PowerCenter resource

Any resource that may be required to run a task. PowerCenter has predefined resources and user-defined resources.

PowerCenter services

The services available in the PowerCenter domain. These consist of the Service Manager and the application services.

primary node

A node that is configured as the default node to run a service process. By default, the Service Manager starts the service process on the primary node and uses a backup node if the primary node fails.

repository domain

A group of linked repositories consisting of one global repository and one or more local repositories.

Repository Service

An application service that manages the PowerCenter repository. It retrieves, inserts, and updates metadata in the repository database tables.

Responder service

A service that invokes a response to an underlying service. An example of a Responder service is an email service that notifies specific users of events. RulePoint contains a number of pre-defined Responder Services.

response

A configurable action that is invoked by specific conditions set by a rule.

rule

Rules are used to analyze events based on specific conditions, and then invoke responses when conditions match. For example, when a service produces an event that matches a specific condition a specific response is invoked.

rule wizard

An easy-to-use application within RulePoint that guides users through each step of rule creation, such as define topics, define conditions, and select responses. The rule wizard then generates the rule.

Service

A service is a configurable program that connects to the outside world and pulls or pushes information into RulePoint or sends out information.

session

A task in a workflow that tells the Integration Service how to move data from sources to targets. A session corresponds to one mapping.

Source service

A service that has a configurable topic and can be scheduled to run at specific times. An example of a source service is a news reader that extracts events from a RSS or Atom news feed. RulePoint contains a number of pre-defined Source Services.

Template

A DRQL rule that uses substitution variables to enable users to create rules from a user interface form.

Topic

A category of events. topics are used to group incoming events into logical categories that are familiar to and defined by users. For example, World News, Transactions, or Stock.

transformation

A repository object in a mapping that generates, modifies, or passes data. Each transformation performs a different function.

Watchlist

Container that stores values as a single object with a unique name that you define. This name then can be referenced in a rule so that the rule can use the data stored in the object. You can modify the values within the watchlist at any time, and any rule referencing that watchlist will use those new values. For example, if you want to create several rules regarding your stock portfolio, you can create a watchlist containing symbols for all of the stocks that you currently own. When you create your rules, you would reference the watchlist instead of specifying each individual stock symbol in multiple rules. In the future, if your portfolio changes, you would simply modify the watchlist instead of individual rules.

Web Services Hub

An application service in the PowerCenter domain that uses the SOAP standard to receive requests and send responses to web service clients. It acts as a web service gateway to provide client applications access to PowerCenter functionality using web service standards and protocols.

Web Services Provider

The provider entity of the PowerCenter web service framework that makes PowerCenter workflows and data integration functionality accessible to external clients through web services.

workflow

A set of instructions that tells the Integration Service how to run tasks such as sessions, email notifications, and shell commands.

workflow instance

The representation of a workflow. You can choose to run one or more workflow instances associated with a concurrent workflow. When you run a concurrent workflow, you can run one instance multiple times concurrently, or you can run multiple instances concurrently.

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